7 Steady State AC Analysis

INTRODUCTION OF AC ANALYSIS

- Alternating current refers to a current which reverses in direction after every half cycle.
  Or
- The current that changes polarity at regular intervals.

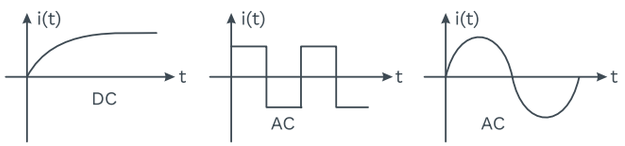


Fig. 7.1 Representation of DC and AC Current

- To judge the ac component in any waveform, we can subtract the average value from the waveform.

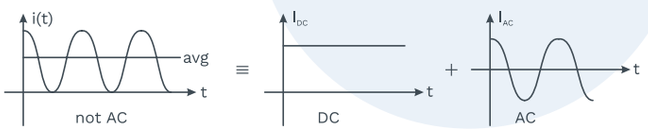


Fig. 7.2 AC Component of Waveform

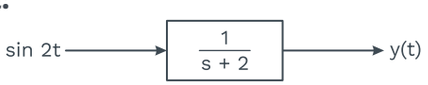
Note: Not all time-varying sources are ac sources.

- Eigen functions are those inputs, when applied to a system, resulting in a similar output.
- For a sinusoidal input, to an LTI system, the output is sinusoid of the same frequency but of different amplitude and phase.
- We generally represent all periodic functions in terms of sinusoids as input & output are the same for sinusoidal function.

Example 1:

Q1. How to determine the output of a system for sinusoidal input?

Sol:



$$H(s) = \frac{1}{s+2} \rightarrow \text{Transfer function}$$

Putting $s = j\omega$

$$H(j\omega) = \frac{1}{j\omega + 2} \rightarrow \text{Frequency response}$$

Grey Matter Alert!

A German-Austrian mathematician and an engineer **Charles Proteus Steinmetz** (1865–1923) pioneered the phasor method in ac circuit analysis.

Definitions

A sinusoid is defined as a signal that has a form of sine or cosine function.

$$\underset{\text{(input)}}{\sin \omega_0 t} \rightarrow \underset{\text{(output)}}{\frac{1}{j\omega_0 + 2} \sin \omega_0 t}$$

$$\sin 2t \rightarrow \frac{1}{j2+2} \sin 2t = \frac{1}{2\sqrt{2} \angle 45°} \sin 2t$$

$$= \frac{1}{2\sqrt{2}} \angle -45 \sin 2t = \frac{1}{2\sqrt{2}} \sin(2t - 45°)$$

- For sinusoidal input, we multiply the magnitude of the transfer function in the amplitude of sine function and shift the phase by an amount equal to the phase of the transfer function.

e.g. $r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$

Points to remember:

- In ac analysis, we use sinusoidal input because it is an eigen function of the LTI system.
- For an LTI system with transfer function H(s) and input $\sin\omega_0 t$, then the output will be -

$$|H(\omega_0)| \sin(\omega_0 t + \angle H(\omega_0))$$

- We always use Fourier transform to analyze the system with sinusoidal input.

SINUSOIDAL STEADY-STATE ANALYSIS

- Steady-state is defined as an equilibrium condition of a circuit or a network that occurs when the effects of the transients are no longer important.
- The final output of a system when transient has died out is called a steady-state response.
- When a sinusoidal signal is applied as an input to a linear circuit, then the circuit produces a steady-state output, which is also a sinusoidal signal. Both the input and the output sinusoidal signals will have the same frequency but different amplitudes and phase angles.



Rack Your Brain

Why the sinusoidal waveform is preferred over any other AC waveform?



Previous Years' Question

In the system shown below, x(t) = sin t. In steady-state, the response y(t) will be:

$$x(t) \longrightarrow \boxed{\frac{1}{s+1}} \longrightarrow y(t)$$

a) $\frac{1}{\sqrt{2}} \sin\left(t - \frac{\pi}{4}\right)$

b) $\frac{1}{\sqrt{2}} \sin\left(t + \frac{\pi}{4}\right)$

c) $\frac{1}{\sqrt{2}} e^{-t} \sin t$

d) sin t – cost

MODELING OF NETWORK ELEMENTS IN FREQUENCY DOMAIN

RESISTANCE(R)

$v(t) = R\, i(t)$

$V(j\omega) = RI(j\omega)$

- Resistance (R) remains the same in the time domain.

INDUCTANCE(L)

$$v(t) = L\frac{di(t)}{dt}$$

$$V(j\omega) = L\left[j\omega I(j\omega)\right]$$

$V(j\omega) = j\omega L\, I(j\omega)$

- Inductance is replaced by $(j\omega L)$

CAPACITANCE(C)

$$i(t) = C\frac{dv(t)}{dt}$$

$I(j\omega) = C[j\omega V(j\omega)]$

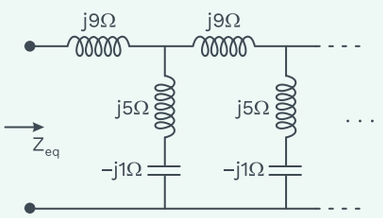
$I(j\omega) = j\omega CV(j\omega)$

$$V(j\omega) = \frac{I(j\omega)}{j\omega C}$$

- C is replaced by $\frac{1}{j\omega C}$

Previous Years' Question

The equivalent impedance Z_{eq} for the infinite ladder circuit shown in the figure is:



a) j12 Ω **b)** –j12 Ω
c) j13 Ω **d)** 13 Ω

Note: In the frequency domain, the relationship between V and I is linear.

IMPEDANCE(Z)

- When inductor and capacitor are used in the frequency domain, they start behaving as a resistor which means they also oppose the flow of ac current.
- The opposition offered by a network to the flow of ac current is termed as impedance.
- Although the impedance is the ratio of phasor voltage V to the phasor current I, it is not a phasor because it does not correspond to a sinusoidally varying quantity.

$$Z = \frac{V}{I}\Omega$$

The impedance of the resistance $Z = R\ \Omega$

The impedance of inductor $Z = j\omega L\ \Omega$

The impedance of capacitor $Z = \frac{1}{j\omega C}\Omega$

- In series connection, $Z_{eq} = Z_1 + Z_2 + ...+ Z_n$
- In a parallel connection, $\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + ... + \frac{1}{Z_n}$
- Impedance Z is a complex number, and thus it follows all the rules defined for a complex number.

Definitions

The admittance is the inverse of impedance, and it measures the flow of current which is allowed by the circuit.

$$Y = \frac{1}{Z} = Z^{-1}$$

ADMITTANCE(Y)

- Admittance represents the allowance to the flow of ac current.
- Admittance is the reciprocal of the impedance i.e.,

  $Y = \frac{1}{Z} = \frac{I}{V}\mho$ or Siemens (S)

  Admittance of resistance $Y_R = \frac{1}{R}\mho$

  Admittance of inductor $Y_L = \frac{1}{j\omega L}\mho$

  Admittance of capacitor $Y_C = j\omega C\,\mho$

- In a series connection,

  $\frac{1}{Y_{eq}} = \frac{1}{Y_1} + \frac{1}{Y_2} + ... + \frac{1}{Y_n}$

- In parallel connection, $Y = Y_1 + Y_2 + ...+ Y_n$

REACTANCE(X)

- Reactance is defined for reactive elements such as L and C, and it represents their reaction when the ac voltage is applied.
- It is the magnitude of impedance for the inductor and capacitor.

For inductor, $X_L = |j\omega L| = \omega L$

For capacitor, $X_C = \left|\frac{1}{j\omega C}\right| = \frac{1}{\omega C}$

Grey Matter Alert!

Susceptance (S) and conductance (G) are real numbers, whereas admittance is a complex number.

- For example: If R and L are connected in series, then impedance is:

$$Z = R + j\omega L = R + jX_L$$

where,

R = Re(Z)

X = Im(Z)

- Real part of the impedance is resistance, and the imaginary part is called reactance.

Note: |j| = 1, |ab| = |a| |b|

SUSCEPTANCE (B)

- Susceptance is defined as inductance and capacitance.
- It represents the allowance to the flow of ac current by inductor and capacitor.
- The imaginary part of admittance is called as susceptance (only magnitude is considered).
- Unit of susceptance is ℧ or Siemens.

$$\text{If, } Z = j\omega L \; ; Y = \frac{1}{j\omega L}; \; B_L = \frac{1}{\omega L}$$

$$\text{If, } Z = \frac{1}{j\omega C} \quad Y = j\omega C \Rightarrow B_C = \omega C$$

Impedance, Z = R + jX

R = Resistance

X = Reactance

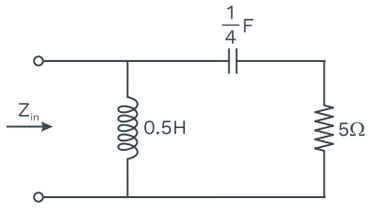
Admittance Y = G + jB

G = Conductance

B = Susceptance

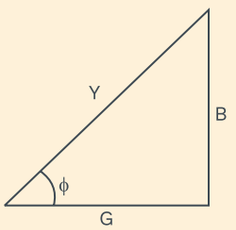
Example 2:

Consider the circuit shown below, find the input impedance Z_{in} (assume ω = 10 rad/s).



Definitions

The admittance triangle is formed by admittance (Y), susceptance (B), and conductance (G) as shown in the graph below:



where,

$$\tan\phi = \frac{\text{Susceptance}}{\text{Conductance}} = \frac{B}{G}$$

Power factor $\cos\phi = \frac{G}{Y}$

Grey Matter Alert!

Difference between impedance and reactance.

Reactance is analogous to resistance for L and C, but it is a real number, whereas impedance is a complex number that can have a real and imaginary part.

Sol:

The impedance of inductor,

$$Z_L = j\omega L = j\times 10\times 0.5 = j5\Omega$$

The impedance of capacitor,

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j\times 10\times \frac{1}{4}} = -j0.4\Omega$$

Since resistor and capacitor are connected in series and their combination is in parallel with the inductor, so the input impedance is:

$$Z_{in} = (5 - j0.4) \parallel j5$$

$$Z_{in} = \frac{(5 - j0.4)\times j5}{5 + j4.6}$$

$$Z_{in} = \frac{(j25 + 2)}{(5 + j4.6)}$$

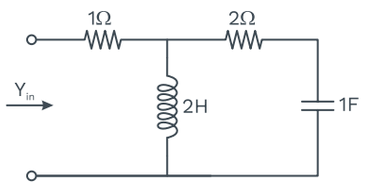
$$Z_{in} = \frac{(2 + j25)(5 - j4.6)}{5^2 + (4.6)^2}$$

$$Z_{in} = \frac{125 + j115.8}{25 + 21.16}$$

$$Z_{in} = \left(2.708 + j2.508\right)\ \Omega$$

Example 3:

Obtain the input admittance (Y_{in}) in the circuit at ω = 1 rad/s.



Sol:

Reactance of inductor,

$$X_L = \omega L = 1 \times 2 = 2$$

$$X_C = \frac{1}{\omega C} = 1$$

From the circuit, 2Ω and 1F are connected in series, so the impedance is,

$$Z_1 = 2 - jX_C = (2 - j)\Omega$$

Rack Your Brain

The inductive susceptance is measured in:

a) Farads **b)** Ohms
c) Henrys **d)** Siemens

The combination is in parallel with 2H, so the impedance is:

$$Z_2 = \frac{(2-j)\cdot j2}{2-j+j2}$$

$$Z_2 = \frac{(2-j)\,j2\,(2-j)}{(2+j)(2-j)} = \frac{\left[4-1-4j\right]j2}{5} = \frac{j6+8}{5}\,\Omega$$

Now, the input impedance is:

$Z_{in} = 1 + Z_2 = (2.6 + j\,1.2)\Omega$

Thus, the input admittance is,

$$Y_{in} = \frac{1}{2.6+j1.2} = \frac{2.6-j1.2}{6.76+1.44} = \frac{26}{82} - j\frac{12}{82}$$

$$Y_{in} = \frac{13}{41} - j\frac{6}{41}$$

$Y_{in} = (0.317 - j0.146)$ ℧

COMPLEX NUMBERS

- A complex number is a mathematical quantity used to express the amplitude and phase shift at once.
- Complex number is represented as:
  $z = x+jy$
- Polar form of a complex number is:
  $z = z\angle\theta^{\circ}$
  $z = |z|\,e^{j\theta}$
  $|z| = \sqrt{x^2+y^2}$

where θ is the argument of z

Magnitude and phase of a complex number:

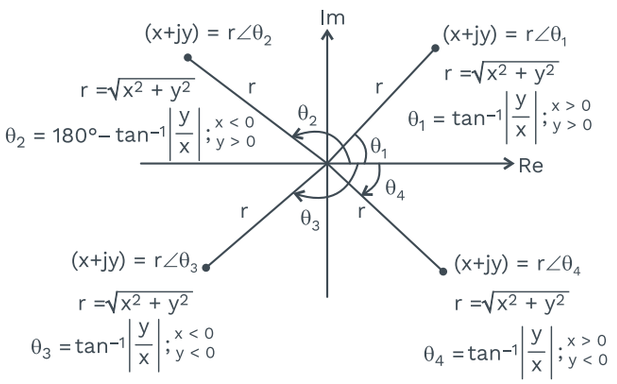
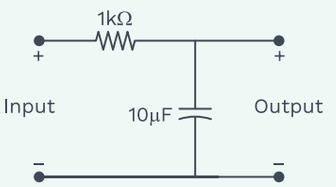


Fig. 7.3 Magnitude and Phase Representation of Complex Number

Previous Years' Question

In the figure, the steady-state output voltage corresponding to the input voltage (3 + 4 sin 100t) V is:



a) $3 + \frac{4}{\sqrt{2}}\sin\left(100t - \frac{\pi}{4}\right)$ V

b) $3 + 4\sqrt{2}\sin\left(100t - \frac{\pi}{4}\right)$ V

c) $\frac{3}{2} + \frac{4}{\sqrt{2}}\sin\left(100t + \frac{\pi}{4}\right)$ V

d) $3 + 4\sin\left(100t - \frac{\pi}{4}\right)$ V

Important formula:

Euler's theorem:

1) $(z)^n = (re^{j\theta})^n = r^n e^{j\theta n}$

2) $r_1\angle\theta_1 \cdot r_2\angle\theta_2 = r_1 r_2\angle\theta_1 + \theta_2$

3) If $z = r\angle\theta$

then, $\frac{1}{z} = \frac{1}{r\angle\theta} = \frac{1}{r}\angle -\theta$

4) $\frac{r_1\angle\theta_1}{r_2\angle\theta_2} = \frac{r_1}{r_2}\angle(\theta_1 - \theta_2)$

5) $(r\angle\theta)^n = r^n\angle n\theta$

Example 4:

Solve the given complex number.

$$\frac{5\angle 30° + (2 - j4)}{(2 + j4)(4 - j5)^*}$$

Sol:

$$= \frac{5[\cos(30) + j\sin(30)] + (2 - j4)}{(2 + j4)(4 + j5)}$$

$$= \frac{(6.33 - j1.5)}{-12 + j26}$$

$$= \frac{(6.33 - j1.5)(-12 - j26)}{(-12 + j26)(-12 - j26)}$$

$$= \frac{(6.33 - j1.5)(-12 - j26)}{(12)^2 + (26)^2}$$

$$= -0.14 - j0.178$$

PHASORS

- A phasor is a complex number representing a sinusoid's amplitude and phase.
- Phasor is a vector representation of the sinusoidal signal to simplify the algebraic operations on sinusoidal quantity.
- Phasors offers an easy method of analyzing linear circuits excited by the sinusoidal sources.
- Sinusoids are easily expressed in terms of phasors.

Grey Matter Alert!

Rectangular and polar are the two forms of complex numbers that are used to represent the phasor quantities.

- Phasor representation of sinusoidal voltage is, $\vec{V} = A\sin(\omega t + \phi)$
- The direction of the phasor represents the phase of sinusoidal quantity, whereas the direction of vector shows its actual direction.

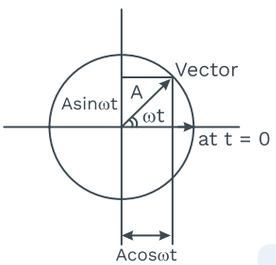


Fig. 7.4 Representation of Vector

Rack Your Brain

The quadrant in which the complex number 24 – j8 is located in the complex plane is:
a) First quadrant
b) Fourth quadrant
c) Second quadrant
d) Third quadrant

- The laws of vector algebra apply to the quantities which are represented as phasors.
- Phasors rotate in anti-clockwise direction at a speed corresponding to their frequency (ω in rad/sec).
- Phasors must have the same frequency when drawing on the same axis.
- If two phasors have different ω means they rotate at different speeds and hence the angle between them continuously changes; thus they cannot be added.

Phasor representation:

Let $c(t) = A\sin(\omega t + \phi)$

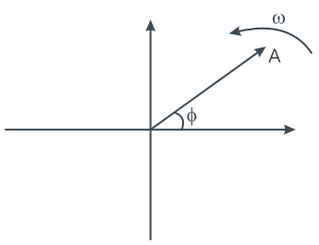


Fig. 7.5 Phasor Representation

- The above phasor diagram represents that the phasor rotates anti-clockwise at a speed of "ω".
- Alternate method of the phasor representation uses rms value of sinusoid as the magnitude of vector as shown in the figure below.

Definitions

Phasors are an efficient method of analyzing the AC circuits when the frequencies are the same. In the phasor diagram, everything is defined relative to the reference phasor.

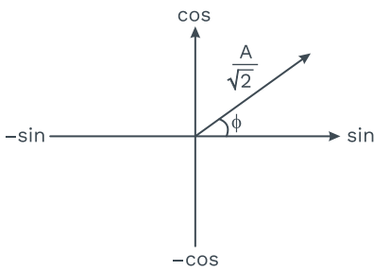


Fig. 7.6 Alternate Method of the Phasor Representation

Rack Your Brain

Which of the following statement is correct regarding a phasor-
i) It may be scalar or vector
ii) It is a time-dependent quantity
iii) It is a complex quantity
a) i)
b) i) and iii)
c) ii) and iii)
d) iii)

- For the laws of vector algebra to apply to phasors, the two phasors must have the same ω or same frequency so that the angle between the two phasors remains constant.
- To represent the two sinusoidal quantities as phasors on the same axis then they both must be in sine or cosine.

Example 5:
Draw the phasor diagram for V = sin (ωt + 30)V and I = cos(ωt – 30)A

Sol:
Given,
V = sin (ωt + 30)V
Using sine as a reference.
I = cos(ωt – 30) = sin(ωt – 30+ 90)
I = sin (ωt + 60)A

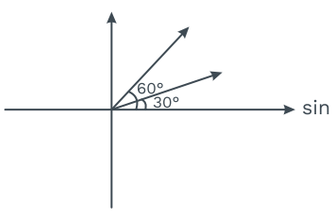


Fig. 7.7

Rack Your Brain

Is phasor a rotating vector?

The necessary condition for representation of phasor:
- Frequency must be the same.
- Both the quantities must be either in sine or cosine.
- The phasor that lies in an anticlockwise direction to the other phasor is said to be leading.
- The phasor that lies in the clockwise direction to the other phasor is said to be lagging.

IMPORTANT RULE

Parallelogram law:

$$\left|\vec{r_1}+\vec{r_2}\right| = [r_1^2 + r_2^2 + 2r_1r_2\cos\theta]^{1/2}$$

$$\left|\vec{r_1}-\vec{r_2}\right| = [r_1^2 + r_2^2 - 2r_1r_2\cos\theta]^{1/2}$$

Phasor addition:

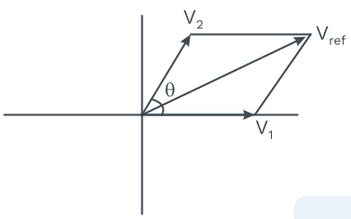


Fig. 7.8

By using parallelogram law:

$$V_{res} = \left[V_1^2 + V_2^2 + 2V_1V_2\cos\theta\right]^{\frac{1}{2}} V$$

Note: V_1 and V_2 must have the same frequency.

Grey Matter Alert!

Phasor analysis can only be applied to the sinusoids.

Phasor diagrams for different elements:

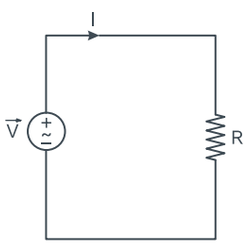
Resistance(R):

- In time domain:

If $v(t) = V_m \sin\omega t$

$$\vec{V} = \frac{V_m}{\sqrt{2}}\angle 0°$$

$$i(t) = \frac{v(t)}{R} = \frac{V_m}{R}\sin\omega t$$



$$\vec{I} = \frac{V_m}{\sqrt{2}\,R}\angle 0°$$

Rack Your Brain

What does the length of the phasor represents?

- So, the phasor diagram of current and voltage for resistance lies in the same phase, i.e., there is no phase shift between v and i.

- In frequency domain:

$Z = R$

$\vec{V} = V_m \angle 0°$

$\vec{I} = \frac{\vec{V}}{Z}$

$\vec{I} = \frac{V_m \angle 0°}{R}$



- If we use the maximum value of the source to represent the phasor then whatever the quantity is derived in the network will always have the maximum value at its magnitude.

Inductance(L):

- In time domain:

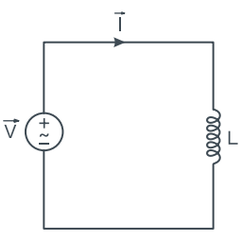
$i(t) = I_m \sin\omega t$

$v(t) = L\frac{d}{dt} i(t)$

$v(t) = L\frac{d(I_m \sin\omega t)}{dt}$

$= \omega L\, I_m \cos\omega t$

$v(t) = \omega L\, I_m \sin(\omega t + 90)$



- In frequency domain:

$\vec{I} = I_m \angle 0°$

$\vec{V} = \vec{I}Z$

$Z = j\omega L$

$\vec{V} = (I_m \angle 0°)(j\omega L)$

$= (I_m \angle 0°)(\omega L \angle 90°)$

$\vec{V} = \omega L I_m \angle 90°$

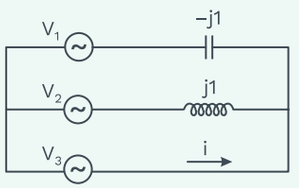
Rack Your Brain

The result of the two alternating sinusoidal voltages or currents can be observed using:

a) Parallelogram law
b) Triangular law
c) Both A and B
d) None of the above

Previous Years' Question

In the given network $V_1 = 100\angle 0$ V, $V_2 = 100\angle -120°$V, $V_3 = 100\angle +120°$V. The phasor current, i (in ampere) is:



a) $173.2\angle -60°$
b) $173.2\angle 120°$
c) $100.0\angle -60°$
d) $100.0\angle 120°$

- $\vec{I}$ lags $\vec{V}$ by 90° or $\vec{V}$ leads $\vec{I}$ by 90°

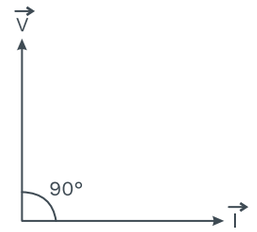


Fig. 7.9 Phasor Diagram

Capacitance(C):

- In time domain:

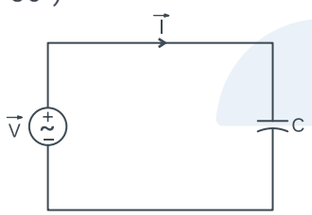
$v(t) = V_m \sin \omega t$

$$i(t) = C\frac{dv(t)}{dt}$$

$$i(t) = C\frac{d(V_m \sin \omega t)}{dt}$$

$= C[\omega V_m \cos \omega t]$

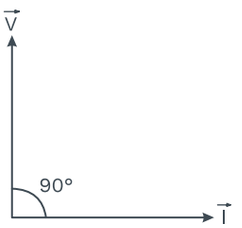
$= \omega C\, V_m \sin(\omega t + 90°)$



$\vec{V} = V_m \angle 0°$

$\vec{I} = \omega C\, V_m \angle 90°$

- $\vec{I}$ leads $\vec{V}$ by 90°



- In frequency domain:

$\vec{V} = V_m \angle 0°$

$$\vec{I} = \frac{\vec{V}}{Z} = \frac{\vec{V}}{\frac{1}{j\omega C}} = j\omega C\,\vec{V}$$

$\vec{I} = (\omega C \angle 90°) \cdot V_m \angle 0°$

$\vec{I} = \omega C V_m \angle 90°$

Important points:

1) For a resistance, current and voltage have the same phase $|I| = \frac{|V|}{R}$

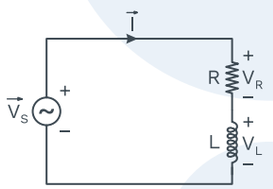
2) For an inductor current lags voltage by 90° $|I| = \frac{|V|}{\omega L}$

3) For a capacitor, current leads voltage by 90° $|I| = \omega C\ |V|$

SERIES CIRCUITS

- In a series circuit, the current is the same in all the elements, so current is used as a reference while drawing the phasor.

1) Series RL circuit:



RL Circuit

- In a series circuit, the current $\vec{I}$ is taken as a reference.

$i(t) = I_m \sin \omega t$

$\vec{I} = I_m \angle 0°$

- For passive element, choose polarity of the voltage steady state current goes from positive (+ve) to negative (–ve) terminal or steady state element absorbs power.

$\vec{V}_R = \vec{I}R$

$\vec{V}_R = (I_m \angle 0°) \times R = I_m R \angle 0°$

$\vec{V}_L = \vec{I}Z_L$

$\vec{V}_L = \vec{I}(j\omega L) = \omega L I_m \angle 90°$

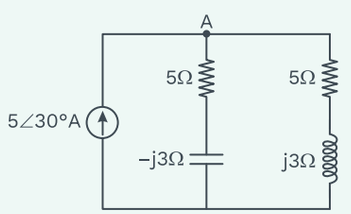
By applying KVL in the circuit,

$\vec{V}_S = \vec{V}_R + \vec{V}_L$

- In ac circuits, voltages are added as phasors, not just by their magnitudes.

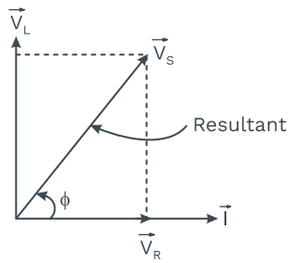
Previous Years' Question

In the AC network shown in the figure, the phasor voltage V_{AB} (in volts) is:



a) 0 V **b)** $5\angle 30°$V

c) $12.5\angle 30°$V **d)** $17\angle 30°$V



Phasor Diagram of RL Circuit

$$|V_S| = \sqrt{V_R^2 + V_L^2}$$

$$|V_S| = \sqrt{(I_m R)^2 + (I_m \omega L)^2}$$

$$|V_S| = I_m \sqrt{(R)^2 + (\omega L)^2}$$

$$\phi = \tan^{-1} \frac{V_L}{V_R}$$

$$\phi = \tan^{-1} \left(\frac{\omega L I_m}{R I_m} \right)$$

$$\phi = \tan^{-1} \left(\frac{\omega L}{R} \right)$$

- $\vec{V}_S$ leads $\vec{I}$ by an angle ϕ

Impedance

$$Z = R + j\omega L$$

$$Z = |Z| \angle \theta$$

$$|Z| = \sqrt{R^2 + \omega^2 L^2}$$

$$\theta = \tan^{-1} \frac{\omega L}{R}$$

$$\vec{I} = I_m \angle 0°$$

$$\vec{V}_S = \vec{I} Z$$

$$\vec{V}_S = (I_m \angle 0°)(|Z| \angle \theta)$$

$$|V_S| = I_m \sqrt{R^2 + \omega^2 L^2}$$

$$\angle V_S = \theta = \tan^{-1} \frac{\omega L}{R}$$

Rack Your Brain

If the phasors are used to represent the maximum values instead of the rms values, then the effect on the phase angle between the quantities is:

a) Increases
b) Remains unchanged
c) Decreases
d) Becomes zero

Series RC circuit:

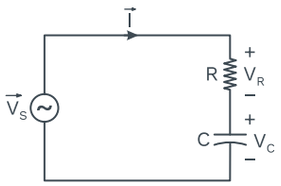


Fig. 7.10 RC Circuit

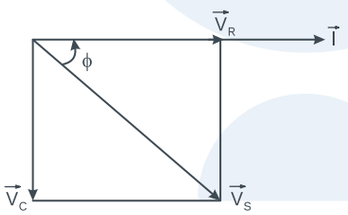
- We assume current as reference.

$\vec{V}_R$ lies in the same phase (ϕ) as $\vec{I}$

$\vec{V}_C$ lags $\vec{I}$ by 90°

$$\vec{V}_C = \vec{I} \times \frac{1}{j\omega C} = I \angle 0 \times -\frac{j}{\omega C}$$

$$\vec{V}_C = \vec{I} \times \frac{1}{j\omega C} = \vec{I}(-jX_C)$$



Phasor Diagram of RC Circuit

- $\vec{I}$ leads $\vec{V}_S$ by an angle ϕ

$$\phi = \tan^{-1}\frac{V_C}{V_R}$$

$$= \tan^{-1}\left(\frac{IX_C}{IR}\right)$$

$$= \tan^{-1}\left(\frac{1}{\omega CR}\right)$$

Impedance,

$$Z = R + \frac{1}{j\omega C}$$

$$= R - jX_C$$

$$= \sqrt{R^2 + X_C{}^2} \angle -\tan^{-1}\frac{X_C}{R}$$

$$Z = |Z| \angle -\theta$$



Rack Your Brain

When a sinusoidal voltage is applied across an R-L series circuit with R = X_L, then the phase angle will be:

a) 90° leading
b) 90° lagging
c) 90°
d) 45° lagging

$$|Z| = \sqrt{R^2 + X_C{}^2}$$

$$\theta = \tan^{-1}\left(\frac{X_C}{R}\right)$$

$$\vec{V}_S = \vec{I}Z$$

$$\vec{V}_S = (I\angle 0°)(|Z|\angle -\theta)$$

$$= I|Z|\angle -\theta$$

$$\angle I = 0$$

$$\angle V_S = \angle -\theta$$

- $\vec{I}$ leads $\vec{V}_S$ by θ.

$$|V_S| = I|Z| = I\sqrt{R^2 + X_C{}^2}$$

$$\angle V_S = -\theta = -\tan^{-1}\left(\frac{X_C}{R}\right)$$

$$\therefore \angle V_S = -\tan^{-1}\left(\frac{1}{\omega RC}\right)$$

Series RLC:

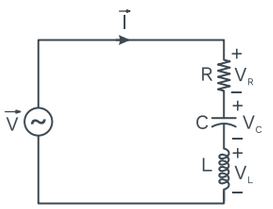
- Consider current as a reference $\left(\vec{I} = I\angle 0\right)$

$$\vec{V}_R = \vec{I}R = IR\angle 0$$

$$\vec{V}_L = \vec{I}\left(j\omega L\right) = I\angle 0 \cdot \left(\omega L\angle 90\right)$$

$$= \omega LI\angle 90°$$

$$\vec{V}_C = \vec{I}\left(\frac{1}{j\omega C}\right) = I\angle 0\left(\frac{1}{\omega C}\angle -90\right) = \frac{I}{\omega C}\angle -90°$$



RLC Circuit

- $\vec{V}_R$ lies in the same phase as $\vec{I}$, $\vec{V}_L$ leads $\vec{I}$ by 90° and $\vec{V}_C$ lags $\vec{I}$ by 90°

Case 1

$X_L > X_C \Rightarrow V_L > V_C$

$V_L = IX_L$

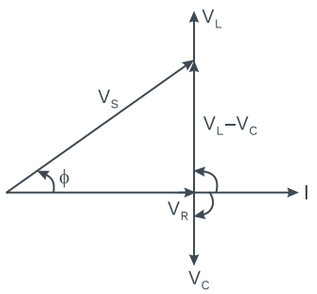
$V_C = IX_C$

Rack Your Brain

For a series RC circuit, if the frequency and the resistance are halved, then the impedance is:

a) Halved
b) Reduced to one-fourth
c) Doubles
d) Cannot be determined

$$\omega L > \frac{1}{\omega C}; \omega > \frac{1}{\sqrt{LC}}$$



Phasor Diagram of RLC Circuit

- V_L and V_C are opposite to each other, and hence when they are added they are subtracted.

$$\bar{Z} = R + j\omega L + \frac{1}{j\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

In this case, $\omega L > \frac{1}{\omega C}$

$$\bar{Z} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \angle \tan^{-1}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

$$\bar{V}_S = \bar{I}\bar{Z} = I\angle 0 \cdot \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \angle \tan^{-1}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

$$= I\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \angle \tan^{-1}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

- $\vec{V}_S$ leads I by $\tan^{-1}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$

Case 2

$X_L < X_C \Rightarrow V_L < V_C$

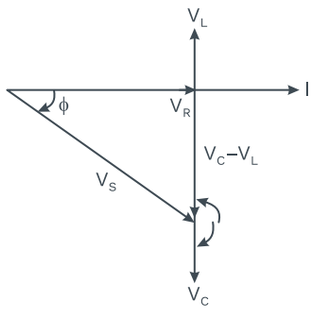
$$\omega L < \frac{1}{\omega C}; \omega < \frac{1}{\sqrt{LC}}$$

Since $V_C > V_L$ resultant of V_C and V_L lies towards V_C.

Previous Years' Question

A DC voltage source is connected across a series RLC circuit. Under steady-state conditions, the applied DC voltage drops entirely across the:

a) R only
b) L only
c) C only
d) R and L combination



Phasor Diagram of RLC Circuit

From phasor diagram V_S lags I by ϕ

$$\bar{Z} = R + j\omega L - \frac{j}{\omega C} = R - j\left(\frac{1}{\omega C} - \omega L\right); \frac{1}{\omega C} > \omega L$$

$$Z = \sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2} \angle - \tan^{-1}\left(\frac{\frac{1}{\omega C} - \omega L}{R}\right)$$

$$\vec{V}_S = \vec{I}Z$$

$$= I\angle 0 \cdot \sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2} \angle - \tan^{-1}\left(\frac{\frac{1}{\omega C} - \omega L}{R}\right)$$

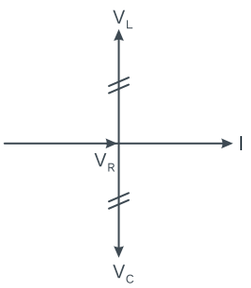
$$= I\sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2} \angle - \tan^{-1}\left(\frac{\frac{1}{\omega C} - \omega L}{R}\right)$$

$\vec{V}_S$ lags $\vec{I}$ by $\tan^{-1}\left(\frac{X_C - X_L}{R}\right)$ or $\vec{I}$ leads $\vec{V}_S$ by $\tan^{-1}\left(\frac{X_C - X_L}{R}\right)$

Case 3

$X_L = X_C \Rightarrow V_L = V_C$

$$\omega L = \frac{1}{\omega C} \text{ or } \omega = \frac{1}{\sqrt{LC}}$$



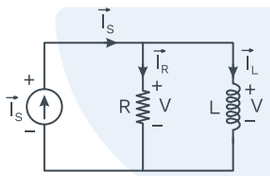
Phasor Diagram for RLC Circuit

- Resultant of V_L and $V_C = 0$
- $\vec{V}_S = \vec{V}_R = \vec{I}\,R$

$\therefore$ $\vec{V}_S$ and $\vec{I}$ in the same phase, and the circuit behaves as the purely resistive circuit.
This condition is called resonance.

Parallel circuits:
In parallel circuits, as the voltage is the same across all the elements so, the voltage is taken as a reference.

Parallel RL circuit:



Parallel RL Circuit

Voltage is the same across both the elements.
I_R lies in the same phase as voltage.

$$\vec{I}_R = \frac{\vec{V}}{R} = \frac{V}{R}\angle 0$$

$\vec{I}_L$ lags $\vec{V}$ by 90°

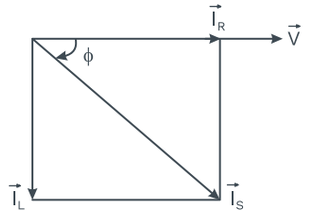
$$\vec{I}_L = \frac{\vec{V}}{j\omega L} = \frac{V\angle 0}{\omega L} \times (-j)$$

$$= \frac{V\angle 0° \times 1\angle -90°}{\omega L}$$

$$\vec{I}_L = \frac{V}{\omega L}\angle -90°$$

By applying KCL in the circuit,

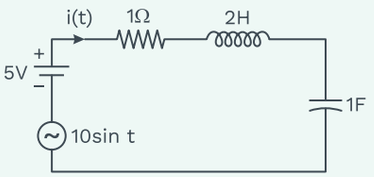
$$\vec{I}_S = \vec{I}_R + \vec{I}_L$$



Phasor Diagram of parallel RL Circuit

Previous Years' Question

In the circuit shown below, the current i(t) under steady-state is:



a) Zero
b) 5
c) 7.07 sin t
d) 7.07 sin(t – 45°)

- From the phasor diagram $\vec{I}_S$ lags $\vec{V}$ by an angle ϕ.

$$\phi = \tan^{-1}\frac{I_L}{I_R}$$

$$\phi = \tan^{-1}\left(\frac{\frac{V}{\omega L}}{\frac{V}{R}}\right)$$

$$\phi = \tan^{-1}\frac{R}{\omega L}$$

$$|I_S| = \sqrt{I_R^2 + I_L^2}$$

$$|I_S| = \sqrt{\left(\frac{V}{R}\right)^2 + \left(\frac{V}{\omega L}\right)^2}$$

$$|I_S| = V \cdot \sqrt{\frac{1}{R^2} + \left(\frac{1}{(\omega L)^2}\right)}$$

- Admittance,

$$\vec{Y} = \frac{1}{R} + \frac{1}{j\omega L} = \frac{1}{R} - \frac{j}{\omega L}$$

$$= \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{\omega L}\right)^2} \angle - \tan^{-1}\frac{\frac{1}{\omega L}}{\frac{1}{R}}$$

$$\vec{Y} = |Y| \angle -\theta$$

$$|Y| = \sqrt{\frac{1}{R^2} + \frac{1}{(\omega L)^2}}\ ;\ \theta = \tan^{-1}\frac{R}{\omega L}$$

$$\vec{I}_S = \vec{V}Y = V\angle 0 \times |Y| \angle -\theta$$

$$|I_S| = V\sqrt{\frac{1}{R^2} + \frac{1}{(\omega L)^2}}$$

$$\angle I_S = -\theta = -\tan^{-1}\left(\frac{R}{\omega L}\right)$$

$\vec{I}_S$ lags $\vec{V}$ by an angle θ.

Previous Years' Question

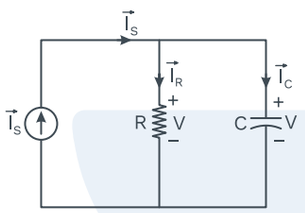
A series RLC circuit consisting of R = 10Ω, $|X_L|$ = 20Ω, and $|X_C|$ = 20Ω, is connected across an ac supply of 200V rms. Then the rms voltage across the capacitor is:

a) $200\angle -90°$ V
b) $200\angle +90°$ V
c) $400\angle +90°$ V
d) $400\angle -90°$ V

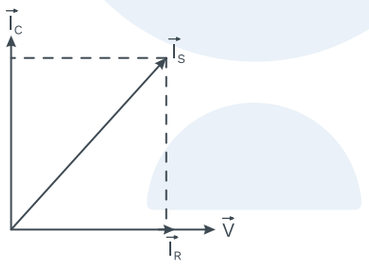
Parallel R-C:

- Consider $\vec{V}$ as reference
- $\vec{I}_R = \frac{\vec{V}}{R} = \frac{V}{R}\angle 0$
- $\vec{I}_C = \frac{\vec{V}}{\frac{1}{j\omega C}} = j\omega C\, V\angle 0$

$= \omega CV \angle 90°$

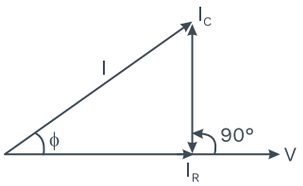


Parallel RC Circuit



Phasor Diagram of Parallel RC Circuit

- I_R lies in the same phase as $\vec{V}$ but I_C leads $\vec{V}$ by 90°



Phasor Diagram for Parallel RC Circuit

By KCL, $\vec{I} = \vec{I}_R + \vec{I}_C$

- $\vec{I}$ leads $\vec{V}$ by ϕ

$$\phi = \tan^{-1}\frac{I_C}{I_R} = \tan^{-1}\left(\frac{\frac{V}{\frac{1}{\omega C}}}{\frac{V}{R}}\right) = \tan^{-1}(\omega RC)$$

$$\bar{Y} = \frac{1}{R} + j\omega C = \sqrt{\frac{1}{R^2} + (\omega C)^2} \angle \tan^{-1}\left(\frac{\omega C}{\frac{1}{R}}\right)$$

$$= \sqrt{\frac{1}{R^2} + (\omega C)^2} \angle \tan^{-1} \omega RC$$

$$\vec{I} = \bar{Y}V$$

$$= \sqrt{\left(\frac{1}{R}\right)^2 + (\omega C)^2} \angle \tan^{-1} \omega RC \cdot V \angle 0$$

$$= V\sqrt{\frac{1}{R^2} + (\omega C)^2} \angle \tan^{-1} \omega RC$$

- $\vec{I}$ leads $\vec{V}$ by an angle $\tan^{-1}(\omega RC)$

Parallel R-L-C:

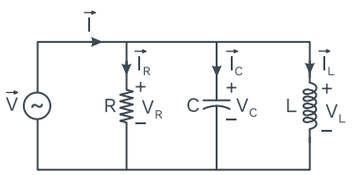
$$\vec{V} = V \angle 0$$

$$\vec{I}_R = \frac{\vec{V}}{R} = \frac{V}{R} \angle 0$$

$$\vec{I}_L = \frac{\vec{V}}{j\omega L} = \frac{V}{\omega L} \angle -90°$$

$$\vec{I}_C = \frac{\vec{V}}{\frac{1}{j\omega C}} = j\omega C \bar{V} = \omega CV \angle 90°$$

- $\vec{I}_R$ lies in the same phase as $\vec{V}$ $\vec{I}_L$ lags $\vec{V}$ by 90° and $\vec{I}_C$ leads $\vec{V}$ by 90°



Parallel RLC Circuit

Case 1

$$X_L > X_C \Rightarrow I_C > I_L;\ \omega > \frac{1}{\sqrt{LC}}$$

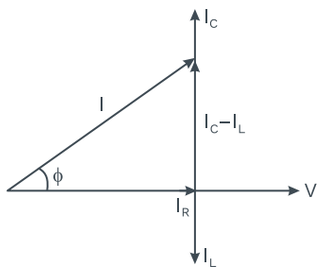
$$I_C = \frac{V}{X_C} \quad I_L = \frac{V}{X_L}$$

Rack Your Brain

When an inductor and a resistor are connected in parallel across a supply, then the current drawn by these two parallel branches will be out of phase by an angle of:

a) 45° **b)** 0°
c) 180° **d)** 90°

If $X_L > X_C$; $I_C > I_L$



Phasor Diagram of Parallel RLC Circuit

Rack Your Brain

The phase angle between the capacitor current and the applied voltage in a parallel RC circuit is:

a) 90° **b)** 180°
c) 45° **d)** 0°

- Since both the currents are opposite, so resultant is the difference of both
- From the phasor diagram $\vec{I}$ leads $\vec{V}$ by ϕ

$$\phi = \tan^{-1}\left(\frac{I_C - I_L}{I_R}\right) = \tan^{-1}\left(\frac{\frac{V}{\frac{1}{\omega C}} - \frac{V}{\omega L}}{\frac{V}{R}}\right)$$

$$\phi = \tan^{-1}\left(\frac{\omega C - \frac{1}{\omega L}}{\frac{1}{R}}\right)$$

Admittance $Y = \frac{1}{R} + \frac{1}{j\omega L} + j\omega C = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$

$$\bar{Y} = \left[\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2\right]^{\frac{1}{2}} \angle \tan^{-1}\left(\frac{\omega C - \frac{1}{\omega L}}{\frac{1}{R}}\right)$$

$$\vec{I} = \vec{V}\,\bar{Y}$$

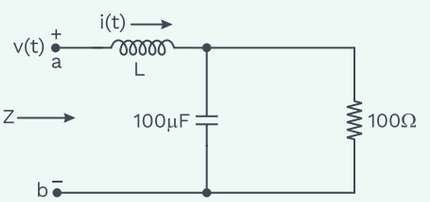
$$= V\angle 0 \cdot \left[\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2\right]^{\frac{1}{2}} \angle \tan^{-1}\left(\frac{\omega C - \frac{1}{\omega L}}{\frac{1}{R}}\right)$$

$$= V\left[\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2\right]^{\frac{1}{2}} \angle \tan^{-1}\left(\frac{\omega C - \frac{1}{\omega L}}{\frac{1}{R}}\right)$$

- $\vec{I}$ leads $\vec{V}$ by an angle ϕ.

Previous Years' Question

The voltage v(t) across the terminals a and b, as shown in the figure is a sinusoidal voltage having a frequency ω = 100 radians/s. When the inductor current i(t) is in phase with the voltage v(t), the magnitude of the impedance Z (in Ω) seen between the terminals a and b is _____ (up to 2 decimal places):

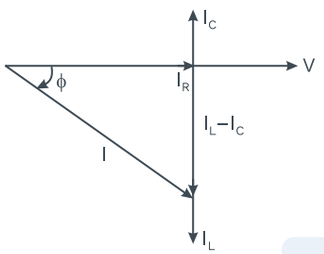


Case 2

$X_L < X_C \Rightarrow I_C < I_L$

$$\frac{V}{X_L} = I_L \; \frac{V}{X_C} = I_C ; X_L < X_C \Rightarrow \omega < \frac{1}{\sqrt{LC}}$$

- Since $I_L > I_C$ resultant of I_L and I_C lies along with I_L.



Phasor Diagram of RLC Circuit

- I lags V by an angle ϕ

$$\phi = \tan^{-1}\left(\frac{I_L - I_C}{I_R}\right) = \tan^{-1}\left(\frac{\frac{V}{\omega L} - \omega CV}{\frac{V}{R}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{1}{\omega L} - \omega C}{\frac{1}{R}}\right)$$

Admittance, $\bar{Y} = \frac{1}{R} + \frac{1}{j\omega L} + j\omega C$

$$= \frac{1}{R} - j\left(\frac{1}{\omega L} - \omega C\right) = \left[\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2\right]^{\frac{1}{2}}$$

$$\angle - \tan^{-1}\left(\frac{\frac{1}{\omega L} - \omega C}{\frac{1}{R}}\right)$$

$\vec{I} = \vec{V}\,Y$

$$= V\angle 0 \cdot \left[\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2\right]^{\frac{1}{2}} \angle - \tan^{-1}\left(\frac{\frac{1}{\omega L} - \omega C}{\frac{1}{R}}\right)$$

$$= V\left[\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2\right]^{\frac{1}{2}} \angle - \tan^{-1}\left(\frac{\frac{1}{\omega L} - \omega C}{\frac{1}{R}}\right)$$

Rack Your Brain

The impedance of parallel RL circuit decreases when the frequency:

a) Increases
b) Decreases
c) Remains constant
d) Is not a factor

$\vec{I}$ lags $\vec{V}$ by $\tan^{-1}\left(\dfrac{\frac{1}{\omega L}-\omega C}{\frac{1}{R}}\right)$

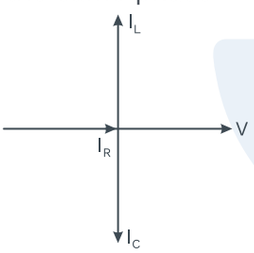
Case 3

$X_L = X_C \Rightarrow I_L = I_C$

resultant of I_L and $I_C = 0$

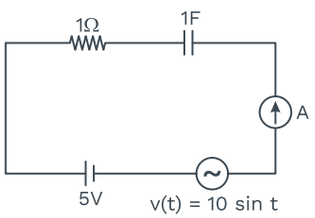
$$\vec{I}_S = \vec{I}_R = \frac{V\angle 0}{R} = \frac{V}{R}\angle 0$$

So circuit behaves as a purely resistive circuit as resistive current and voltage lie in the same phase.



Example 6:

Find reading of ammeter of the circuit shown in the figure:



a) 10 A **b)** 5 A

c) 8 A **d)** 0 A

Sol:

Use superposition theorem,

Capacitor will acts as an open circuit.

$I_{DC} = 0$

For AC, $Z_C = \dfrac{1}{j\times 1\times 1} = -j$

$Z = 1 - j = \sqrt{2}\angle -45°$

$$\vec{I}_{AC} = \frac{\frac{10}{\sqrt{2}}}{\sqrt{2}\angle -45°} = 5\angle 45° \text{ amp}$$

Rack Your Brain

For a parallel RLC circuit, if the current through the capacitor and the inductor is the same, then the power factor is:

a) Zero **b)** Unity

c) Lagging **d)** Leading

Grey Matter Alert!

For a series RLC circuit, when the capacitive reactance is equal to the inductive reactance, then the total reactance is equal to:

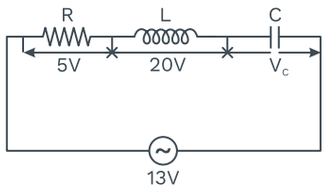
a) Less than one

b) Unity

c) Greater than one

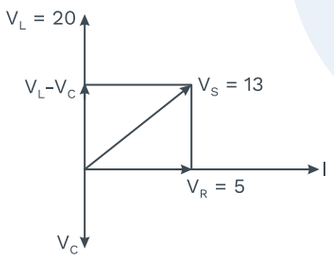
d) Zero

Example 7:
Find the voltage across the capacitor of the circuit shown in the figure:



a) 8V **b)** 32V
c) 12V **d)** 8V or 32V

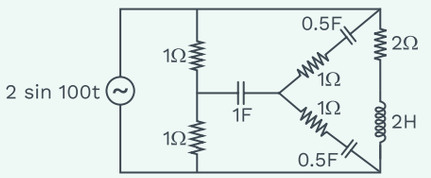
Sol:
For the above circuit, the phasor diagram is:



$13^2 = 5^2 + (V_L - V_C)^2$
$(V_L - V_C)^2 = 12^2$; $|V_L - V_C| = 12$
$V_L - V_C = 12$ $[V_L > V_C]$
$V_C = 20 - 12 = 8$ volt
$V_C - V_L = 12$ $[V_C > V_L]$
$V_C = 20 + 12 = 32$ volt

Previous Years' Question

The value of current through the 1F capacitor as shown in the figure is:



a) 0 **b)** A1
c) 2 **d)** 3

Example 8:
Two circuit elements in a series connection have current and total voltage

$i(t) = 13.42 \sin(500t - 53.4°)A$

$v(t) = 150 \sin(500t + 10°)V$

Identify the two elements and their values.

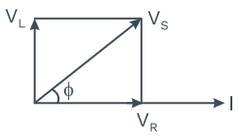
Sol:
Given,

$i(t) = 13.42 \sin(500t - 53.4°)A$

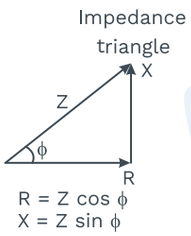
$v(t) = 150 \sin(500t + 10°)V$

i(t) lags v(t) by $10 - (-53.4) = 63.4°$

$$Z = \frac{V}{I} = \frac{150}{13.42} = 11.177\,\Omega$$



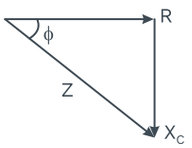
$V_S \cos\phi = V_R$
$IZ \cos\phi = IR$
$R = Z \cos\phi$
and $X = Z \sin\phi$



$R = 11.177 \cos 63.4 = 5\ \Omega$
$X = 11.177 \sin 63.4 = 10\ \Omega$

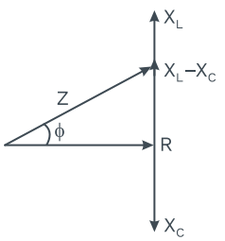
$$L = \frac{X}{\omega} = \frac{10}{500} = 0.02\,H = 20\,mH$$

Series R-C:



RC Circuit

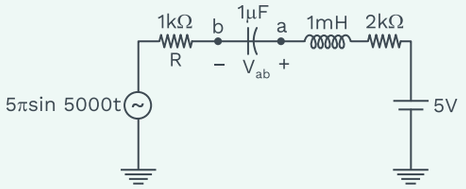
Resistance = Re (Z) = Z cos ϕ
Reactance = Im (Z) = Z sin ϕ
cos ϕ = power factor
Series RLC
$X_L > X_C$



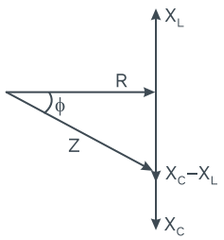
Series RLC Circuit

Previous Years' Question

In the circuit shown, the average value of the voltage V_{ab} (in volts) in the steady-state condition is:



$X_C > X_L$



POWER IN SINUSOIDAL STEADY-STATE

A sinusoidal voltage $v(t) = V_m \cos\omega t$, applied across an impedance, $Z = |Z| \angle\theta$ establishes a current:

$i(t) = I_m \cos(\omega t - \theta)$

$$\vec{V} = \frac{V_m}{\sqrt{2}} \angle 0°$$

$$I = \frac{\vec{V}}{Z} = \frac{\frac{V_m}{\sqrt{2}} \angle 0°}{|Z| \angle\theta}$$

$$I = \frac{V_m \angle -\theta}{\sqrt{2}\,|Z|}$$

$i(t) = I_m \cos(\omega t - \theta)$

and $I_m = \frac{V_m}{|Z|}$

or $|Z| = \frac{V_m}{I_m}$

So, the ratio of voltage and current is termed impedance.

Instantaneous power:

Instantaneous power is defined as the product of the instantaneous terminal voltage and current i.e.,

$p(t) = v(t).i(t)$ Watts

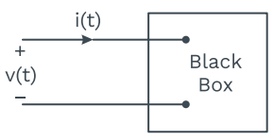


Fig. 7.11

Figure: A black box with instantaneous voltage and instantaneous current.

$i(t) = I_m \cos(\omega t + \theta_i)$

$v(t) = V_m \cos(\omega t + \theta_v)$

ω is same for both the quantities as the network is linear

Note: The instantaneous power assuming the passive sign convention.

- If the current is in the direction of voltage drop (+ –), then the instantaneous power is-

  $p(t) = v(t).i(t)$ Watts

and, if the current is in the direction of voltage rise (– +), then the instantaneous power is-

$p(t) = -v(t).i(t)$ Watts

$p(t) = V_m \cos(\omega t + \theta_v).\ I_m \cos(\omega t + \theta_i)$

$$p(t) = \frac{V_m I_m}{2} [\cos(2\omega t + \theta_v + \theta_i) + \cos(\theta_v - \theta_i)]$$

Where, $\theta_v - \theta_i = \theta$

and θ = angle of impedance (Z)

$$\because \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}}$$

$$\therefore \frac{V_m I_m}{2} = V_{rms} \cdot I_{rms}$$

$$p(t) = V_{rms} I_{rms} \left[\underbrace{\cos(2\omega t + \theta_v + \theta_i)}_{\text{double frequency and time dependent}} + \underbrace{\cos(\theta_v - \theta_i)}_{\text{constant}} \right]$$

Fig. 7.12

Note: Power in ac oscillates at twice the frequency as compared to the source frequency.

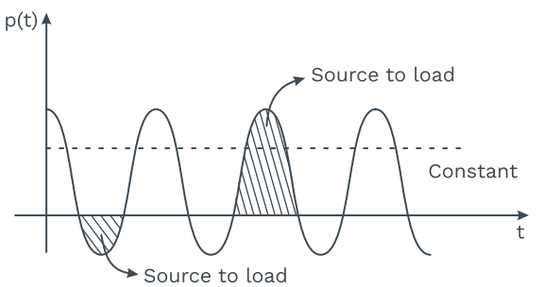


Fig. 7.13 Waveform for Instantaneous Power

Note: When power p(t) is positive, the power is absorbed by the circuit and when power p(t) is negative, power is supplied by the source, as shown in the graph above.

- The instantaneous power changes with time and is, therefore, difficult to measure.
- The average power is more convenient to measure as compared to instantaneous power. In fact wattmeter responds to average power only.

Example 9:
A boiler is fed from a 230V, 50Hz supply. Find the frequency of the instantaneous power of the boiler.

Sol:
We know that the frequency of power is:

f_p = 2 × source frequency

$f_p = 2f_s$

$\therefore f_p = 2 \times 50$

f_p = 100Hz

Example 10:
Calculate the instantaneous power p(t) for a voltage v = 120cosωt V which is connected across an impedance Z = 5∠−60°Ω .

Sol:
Given,

v = 120cosωt V

Z = 5∠−60° Ω

We know, impedance (Z) is:

$$Z = \frac{v}{i}\cos(\omega t - \theta)$$

$$i = \frac{v}{Z}\cos(\omega t + 60°)$$

$$i = \frac{120}{5}\cos(\omega t + 60°)$$

$$i = 24\cos(\omega t + 60°)\ A$$

$$\vec{V} = \frac{120}{\sqrt{2}}\angle 0° = 84.85\angle 0°\ V$$

$$\vec{I} = \frac{\vec{V}}{Z}$$

$$= \frac{84.85\angle 0°}{5\angle -60°}$$

$$\vec{I} = 16.97\angle 60°\ A$$

Rack Your Brain

In a series of RLC circuits when the inductive reactance is greater than the capacitive reactance ($X_L > X_C$) then the circuit is:

a) Impedance in nature
b) Inductive in nature
c) Resistive in nature
d) Capacitive in nature

Therefore, instantaneous power is:

$p(t) = v(t).i(t)$

$= 84.85 \times 16.97\,[\cos(2\omega t + 60°) + \cos 60°]$

$p(t) = 1439.95[\cos(2\omega t + 60°) + \cos 60°]\,W$

Average power or real power:

- Average power is defined as the average of the instantaneous power over one period. The average power or real power is measured in watts.
- The average power corresponding to the instantaneous power p(t) can be obtained by integrating the instantaneous power over one period of the sinusoidal signal.

Therefore, the average power is:

$$P_{avg} = \frac{1}{T}\int_0^T p(t)\,dt$$

$$\underbrace{P_{avg} \times T = \int_0^T p(t)d(t)}_{\text{Energy in one cycle}}$$

Note: An energy transferred from one point to another depends on the average power.

$$P_{avg} = \frac{1}{T}\int_0^T V_{rms} I_{rms}[\underbrace{\cos(2\omega t + \theta_v + \theta_i)}_{\Downarrow\ 0} + \cos(\theta_v - \theta_i)]dt$$

$$\therefore\ P_{avg} = \frac{1}{T}[V_{rms}\, I_{rms} \cos(\theta_v - \theta_i)T]$$

Thus, the average power in an ac circuit is given by-

$\therefore\ P_{avg} = V_{rms}\, I_{rms} \cos(\theta_v - \theta_i)\,W$

Fun facts: Average power takes part in the energy conversion process.

Example 11:

If $v(t) = 110\cos\omega t$ V and impedance $Z = 2\angle -60°\Omega$. Calculate the average power.

Sol:

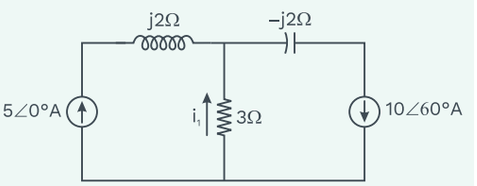
Given,

$v(t) = 110 \cos \omega t$ V

$Z = 2\angle -60° \Omega$

Previous Years' Question

For the circuit as shown in the figure, then the instantaneous current $i_1(t)$ is:



a) $\frac{10\sqrt{3}}{2}\angle 90°$ Amps

b) $\frac{10\sqrt{3}}{2}\angle -90°$ Amps

c) $5\angle 60°$ Amps

d) $-5\angle 60°$ Amps

We know,

$$V_{rms} = \vec{V} = \frac{110}{\sqrt{2}} = 77.78V$$

$$I_{rms} = \vec{I} = \frac{\vec{V}}{Z} = \frac{77.78\angle 0°}{2\angle -60°}$$

$$\vec{I} = 38.89\angle 60° \text{ A}$$

$\therefore \vec{I}$ leads $\vec{V}$ by 60°

The average power is:

$P_{avg} = V_{rms}\, I_{rms} \cos(\theta_v - \theta_i)$
$P_{avg} = 77.78 \times 38.89 \cos(0 - 60°)$
$P_{avg} = 3024.93 \cos 60$

$$P_{avg} = 1512.46 \text{ W}$$

Fun facts: A resistive load (R) absorbs power all the time, whereas a reactive load, i.e. inductor (L) and capacitor (C) absorbs zero average power.

Example 12:
For a passive linear network, if voltage
$v(t) = 60 \cos(377t + 40°)$ V and
$i(t) = 10 \cos(377t - 10°)$ A
Find the instantaneous power and the average power absorbed in the network.

Sol:
Given,
$v(t) = 60 \cos(377t + 40°)$ V
$i(t) = 10 \cos(377t - 10°)$ A
We know that the instantaneous power is:

$p(t) = v(t) \cdot i(t)$
$= 60 \cos(377t + 40°) \cdot 10\cos(377t - 10°)$

$$= \frac{600}{2}[\cos(754t + 30°) + \cos 50°]$$

$= 300[\cos(754t + 30°) + \cos 50°]$

$p(t) = 192.83 + 300 \cos(754t + 30°)$W

The average power is:

$$P_{avg} = \frac{1}{2} V_m \cdot I_m \cos(\theta_v - \theta_i)$$

$$= \frac{1}{2} \times 60 \times 10 \cos[40 - (-10)]$$

$$= 300\cos 50°$$

$$\boxed{P_{avg} = 192.83W}$$

Example 13:
Obtain the average power absorbed by an impedance Z = 20 - j70Ω. When a voltage v = 40∠0° V is applied across it.

Sol:
Given,

$Z = (20 - j70)\Omega$

$v = 40\angle 0°$ V

We know that the current through the impedance is:

$$I = \frac{V}{Z}$$

$$I = \frac{40\angle 0°}{20 - j70}$$

$$I = \frac{40\angle 0°}{72.8\angle -74.05°}$$

$I = 0.549\angle 74.05°$A

Then the average power is:

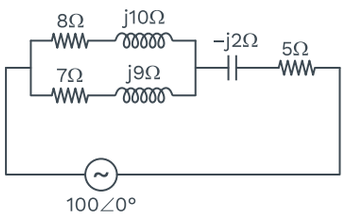
$$P_{avg} = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$= \frac{1}{2} \times 40 \times 0.549 \cos(0 - 74.05°)$$

$P_{avg} = 10.98\cos(-74.05°)$

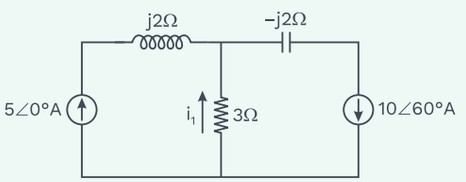
$P_{avg} = 3.017W$

Example 14:
For the circuit shown below. Calculate the average power consumed.



Previous Years' Question

For the circuit in the figure, the instantaneous current i_1 is:



a) $\frac{10\sqrt{3}}{2}\angle 90°$ Amps

b) $\frac{10\sqrt{3}}{2}\angle -90°$ Amps

c) $5\angle 60°$ Amps

d) $5\angle -60°$ Amps

Sol:

Given circuit:

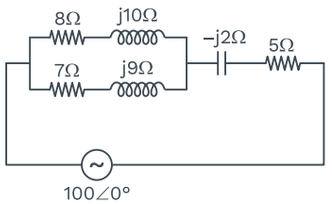


Fig. 7.14

In this circuit, (8 + j10)Ω is in parallel with (7+j9) Ω then,

$Z_1 = (8 + j10) \parallel (7 + j9)$

$$Z_1 = \frac{(8 + j10)(7 + j9)}{(8 + j10) + (7 + j9)} = \frac{(8 + j10)(7 + j9)}{15 + j19}$$

$Z_1 = (3.764 + j4.775)\Omega$

Now, Z_1 is in series with (5 – j2)Ω then:

$Z_{eq} = Z_1 + (5 - j2)$

$= 3.764 + j4.775 + 5 - j2$

$Z_{eq} = (8.764 + j2.77)\Omega$

Current, $I = \frac{V}{Z_{eq}}$

$$I = \frac{100 \angle 0°}{8.764 + j2.77}$$

$$I = \frac{100 \angle 0°}{9.191 \angle 17.540°}$$

$I = 10.88 \angle -17.54°A$

∴ Average power, $P_{avg} = V\, I \cos[0 - (-17.54)]$

$P_{avg} = 100 \times 10.88 \cos(17.54)$

$P_{avg} = 1088 \times 0.9535$

$P_{avg} = 1037.415$ W

Previous Years' Question

The circuit shown in the figure has two sources connected in the series. The instantaneous voltage of the AC source (in volt) is given by v(t) = 12 sin t. If the circuit is in a steady state, then the rms value of the current (in ampere) flowing in the circuit is _____.



Active power:

- Active power is the power that is capable of doing work in the system or capable of energy conversion.
- As studied earlier, there is only one power capable of energy transfer, and that power is average power.
- Hence, active power is nothing but average power, and it is also referred to as real power, i.e. Average power = active power = real power.

  Active power = $V_{rms}\, I_{rms} \cos(\theta_v - \theta_i)$

Behaviour of different elements:

1) Resistance:

Current and voltage lie in the same phase.



Fig 7.15 Phasor Diagram for Resistance

$\therefore$ From the above phasor diagram,

$\theta_v - \theta_I = 0$

$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i) = V_{rms} I_{rms} \cos(0)$

$P = V_{rms} \cdot I_{rms}$

$\because \; I_{rms} = \frac{V_{rms}}{R}$

$\therefore \; P = \frac{V_{rms}^2}{R} = I_{rms}^2 R$

$v(t) = V_m \sin\omega t$ V

$i(t) = I_m \sin\omega t$ I

where, $I_m = \frac{V_m}{R}$

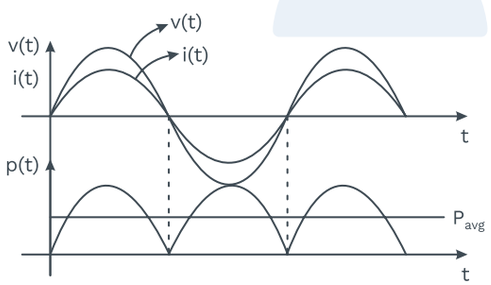


Fig. 7.15 Waveform Representing the Behaviour of Resistance

- In the case of resistance, the instantaneous power is always positive because it cannot return power to the source, but it can only take energy from the source and convert it to heat energy.

2) Inductor:

In an inductor, the current lags the voltage by 90°.

Grey Matter Alert!

Average power is the most important because it represents the energy transferred from one point to another.
For example, Generator converts the mechanical energy input to the electrical energy output and the power involved in this entire process is an average power.

Rack Your Brain

Why average power is preferred over instantaneous power?

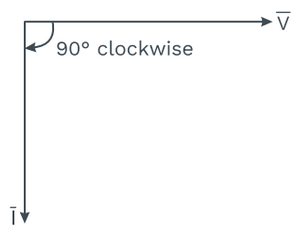


Fig. 7.16 Phasor Diagram for Inductor

From the above phasor:

$\theta_v = 0$

$\theta_i = 90°$ (clockwise)

$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$

$P = V_{rms} I_{rms} \cos(0 - (-90°))$

$P = V_{rms} I_{rms} \cos(90°)$

$P = 0,$

- Since the average power in an inductor is zero, it implies that the inductor does not absorb any energy.

$i(t) = I_m \sin\omega t$

$$v(t) = L\frac{di(t)}{dt} = \omega L I_m \cos\omega t$$

$p(t) = v(t).i(t)$

$$p(t) = \omega L I_m^2 \sin\omega t \cos\omega t$$

$$p(t) = \frac{1}{2}\omega L I_m^2 \sin 2\omega t$$

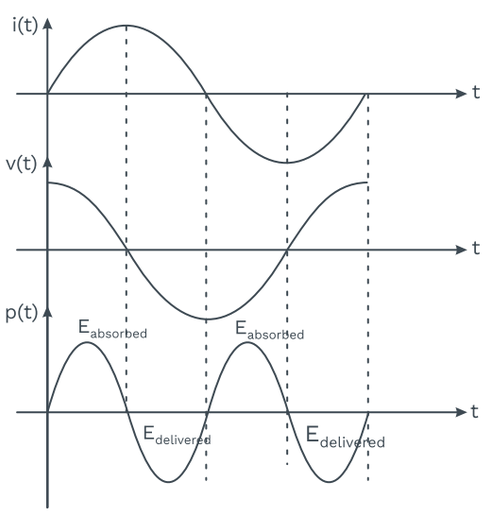


Fig. 7.17 Waveform Representing the Behaviour of an Inductor

Grey Matter Alert!

If average power comes out to be zero, it means that there is no net energy transferred between the source and load. It implies that the amount of energy delivered by the source to the load is the same as the energy returned by the load to the source.

- From the above waveform of instantaneous power, the inductor does not absorb any net electrical energy but whatever energy the inductor absorbs in one-half cycle, it returns in the next half-cycle.
- If a periodic current is applied to an inductor, then the value of current at t = 0 and t = T is the same, and hence the value of energy $\left(\frac{1}{2}Li^2\right)$ is also the same at t = 0 and t=T. So, the inductor does not absorb or deliver any net energy in one cycle.

3) Capacitor:

- In a capacitor, current leads voltage by 90°.

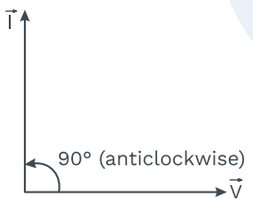


Fig. 7.18 Phasor Diagram for Inductor

From the above phasor:

$\theta_v = 0°$

$\theta_i = 90°$ (anticlockwise)

$P = V_{rms}I_{rms}\cos(\theta_v - \theta_i)$

$= V_{rms}I_{rms}\cos(0 - 90°)$

$P = V_{rms}I_{rms}\cos(90°)$

$P = 0$

- Since, average power is zero for a capacitor so even a capacitor does not absorb or deliver any net energy.

$v(t) = V_m \sin\omega t$

$$i(t) = C\frac{dv}{dt}$$

$i(t) = \omega CV_m \cos\omega t$

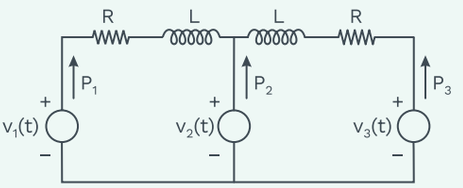
$p(t) = v(t).i(t)$

$= V_m \sin\omega t\ \omega CV_m \cos\omega t$

$= \omega CV_m^2 \sin\omega t.\cos\omega t$

Previous Years' Question

In the figure, the voltages are $v_1(t) = 100\cos(\omega t)$, $v_2(t) = 100\cos(\omega t + \pi/18)$ and $v_3(t) = 100\cos(\omega t + \pi/36)$. The circuit is in sinusoidal steady state, and $R << \omega L$. P_1, P_2 and P_3 are the average power outputs. Which one of the following statements is true?



a) $P_1 = P_2 = P_3 = 0$

b) $P_1 < 0, = P_2 > 0, P_3 > 0$

c) $P_1 < 0, = P_2 > 0, P_3 < 0$

d) $P_1 > 0, = P_2 < 0, P_3 > 0$

$$p(t) = \frac{1}{2}\omega C V_m^2 \sin 2\omega t$$

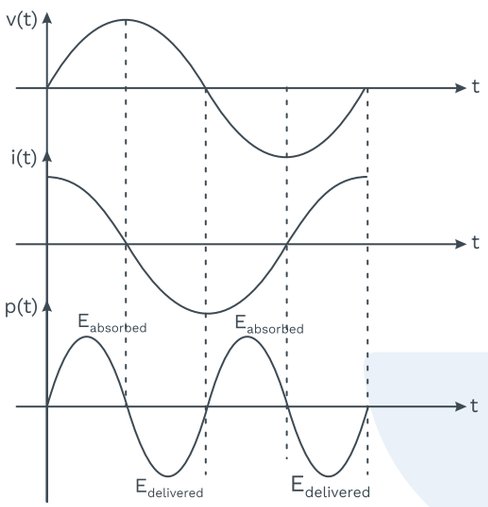


Fig. 7.19 Waveform Representing the Behaviour of Capacitor

- Similar to an inductor, the capacitor also absorbs and delivers energy in every half-cycle so the net energy delivered or absorbed is zero for ac.
- If a periodic function is applied to a capacitor, the net energy absorbed or delivered over one cycle is zero.

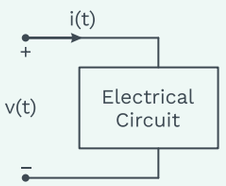
Previous Years' Question

The voltage across the circuit in the figure, and the current through it, are given by the following expressions

$$v(t) = 5 - 10\cos(\omega t + 60°)V$$

$$i(t) = 5 + X\cos(\omega t)A$$

where $\omega = 100\pi$ radian/s. If the average power delivered to the circuit is zero, then the value of X (in ampere) is ____ (up to 2 decimal places).



Note: Inductor and capacitor absorb energy from the source to set up magnetic and electric fields inside them, and when these fields gradually vanished, then this energy is returned.

Apparent power:

- Apparent power is the product of the root mean square (RMS) value of voltage and current.
- Apparent power, $S = V_{rms}\ I_{rms}$.
- It is measured in kVA or MVA.
- In DC circuits, power is simply given as P=VI; for ac circuits, the observer feels that the power in the system must be $V_{rms}.\ I_{rms}$, whereas the power is $V_{rms}I_{rms}\cos(\theta_v - \theta_i)$. The power imagined by the observer which is not present is called apparent power.

Example 15:
A system is designed to handle 100V and 4A, and it is expected the power of the system is to be 400W, but the actual power is $400\cos(\theta_v - \theta_i) < 400$. Why?

Sol:
Due to the phase difference between $\vec{V}$ and $\vec{I}$, we do not receive power as $V_{rms}I_{rms}$.

Important points:

- Real power is not equal to the apparent power because of phase shift between $\vec{V}$ and $\vec{I}$ which is produced due to L and C.
- In the case of resistance, real power is the same as apparent power. The average power P_{avg} is non-zero for resistance, but it is zero for L and C, so we need to define a quantity for L and C which is analogous to average power in resistance.

Fun fact:
Expected power = Apparent power (S)
Actual power = Real power

Power factor:

- The ratio of the real power to the apparent power is called as power factor.
- Power factor represents what fraction of expected or apparent power exists in the system.
- Power factor, $pf = \dfrac{V_{rms} \cdot I_{rms} \cos(\theta_v - \theta_i)}{V_{rms}\, I_{rms}}$

$$pf = \frac{\text{Real power}}{\text{Apparent power}}$$

$$pf = \frac{V_{rms} \cdot I_{rms} \cos(\theta_v - \theta_i)}{V_{rms}\, I_{rms}} = \frac{P}{S}$$

$pf = \cos(\theta_v - \theta_i)$
$pf = \cos\phi < 1$

Note: The power factor also represents how efficiently is the capacity of a plant is utilized. So, it is very clear that the power factor must be as high as possible.

Points to remember:
Power factor $pf = \cos(\theta_v - \theta_i) = \cos\phi$

Grey Matter Alert!

Active power	Reactive power
The active power is the real power that is dissipated in the circuit.	The power which moves back and forth between the load and the source is known as the reactive power.
Active power, $P = V \times I \cos\phi$ $P = V\ I \cos\phi$	Reactive power, $Q = V \times I \sin\phi$ $Q = V\ I \sin\phi$
The unit of active power is Watts, kW, MW.	The unit of reactive power is VAR, kVAR, MVAR.
Active power is used to produce heat in a heater, light in lamps, and torque in the motor.	Reactive power is used to measure the power factor of the circuit.

We do not take the power factor as $\cos(\theta_v - \theta_i)$ because most of the practical sources are voltage sources, and they are considered as reference, i.e. we observe lag or lead of current with respect to the voltage.

Nature of power factor in elements:

1) **Resistance:** In resistance, $\vec{V}$ and $\vec{I}$ lies in the same phase.

   $\therefore \quad \cos(\theta_v - \theta_i) = \cos(0°)$

   $\therefore \quad pf = 1$

   It is called as unity power factor

2) **Inductance:** In inductance, $\vec{I}$ lags $\vec{V}$ by 90° i.e.,

   $\vec{V} = V\angle 0°\ V$

   $\vec{I} = I\angle -90°\ A$

   $\therefore \quad \cos(\theta_v - \theta_i) = \cos(0°\text{-}(\text{-}90°)) = \cos 90°$

   pf = 0 (lagging)

3) **Capacitance:** In capacitance, $\vec{I}$ leads $\vec{V}$ by 90°

   i.e., $\vec{V} = V\angle 0°\ V$

   $\vec{I} = I\angle 90°\ A$

   $\therefore \quad \cos(\theta_v - \theta_i) = \cos(0°\text{-}90°)) = \cos(-90°)$

   Power factor, $pf = \cos(\theta_v - \theta_i) = 0$

   Pf = 0 (leading)

Note: The nature of the power factor (lag or lead) depends on the nature of the load, whether it is inductive or capacitive.

Case I: For RL load

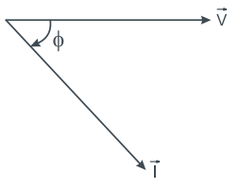


Fig. 7.20 Phasor Diagram of RL Load

In this case, $\vec{I}$ lags $\vec{V}$ by ϕ

$$\phi = \tan^{-1}\left(\frac{\omega L}{R}\right) \text{ : series}$$

$$\phi = \tan^{-1}\left(\frac{R}{\omega L}\right) \text{ : parallel}$$

So, the power factor is:

pf = cosϕ (lagging)

Case II: For RC load

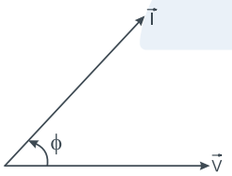


Fig. 7.21 Phasor Diagram of RC Circuit

In this case, $\vec{I}$ leads $\vec{V}$ by ϕ

$$\phi = \tan^{-1}\left(\frac{1}{\omega RC}\right) \text{ : series}$$

$\phi = \tan^{-1}(\omega RC)$: parallel

So, the power factor is:

pf = cosϕ (leading)

Rack Your Brain

Why does the inductor not absorb any energy?

- For example: If $\vec{V} = 95\angle 0°$ V and $\vec{I} = 5$ A at 0.8 power factor lagging, which means:

cosϕ = 0.8

ϕ = 36.86

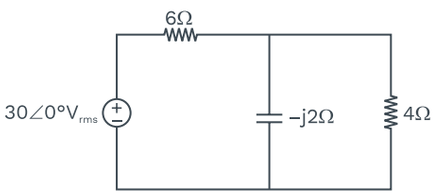
Thus, I lags V by 36.86

$\vec{I} = 5\angle -36.86°$ A

Note: The power factor explains what fraction of total power is useful for the system. In the case of resistance, the power factor is unity, which means the entire power is useful but in the case of inductor and capacitor, the power factor is zero, which means the entire power is not useful.

Important point:
The angle between voltage and current is the same as the angle of impedance or admittance which is also the angle of power factor.
i.e., Power factor angle = Angle between $\vec{V}$ and $\vec{I}$ = Angle of impedance

Example 16:
For the circuit shown below, what is the power factor seen by the source and the average power delivered by the source?



Sol:
In the given circuit, 4Ω is parallel to –j2Ω
So, $Z_1 = 4\Omega \parallel (-j2)\Omega$

$$Z_1 = \frac{4 \times (-j2)}{4 + (-j2)}$$

$$Z_1 = \frac{-j8}{4 - j2}\Omega$$

Now, this Z_1 is in series with 6Ω, then the equivalent impedance is:

$$Z_{eq} = 6 + \frac{(-j8)}{4 - j2}$$

$$Z_{eq} = 6 + \left(\frac{-j8 \times (4 + j2)}{4 - j2 \times (4 + j2)}\right)$$

Previous Years' Question

The average power delivered to an impedance (4 – j3)Ω by a current 5cos(100πt + 100) A is:
a) 44.2 W **b)** 50 W
c) 62.5 W **d)** 125 W

$$Z_{eq} = 6 + \frac{(16 - j32)}{16 + 4}$$

$$Z_{eq} = 6 + \frac{16 - j32}{20}$$

$$Z_{eq} = \frac{120 + 16 - j32}{20} = \frac{136 - j32}{20}$$

$$Z_{eq} = (6.8 - j1.6)\Omega$$

- Since, imaginary part of the impedance is negative.

$\therefore$ Im (Z) < 0; capacitive impedance (Z)

- Thus, it is the leading power factor

Power factor angle (ϕ) = impedance angle (θ)

So, from impedance triangle:

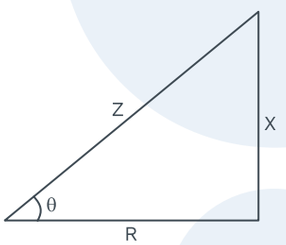


Fig. 7.22 Impedance Triangle

$$\cos\phi = \cos\theta = \frac{R}{Z}$$

$$= \frac{6.8}{\sqrt{(6.8)^2 + (1.6)^2}}$$

Power factor, cos θ = 0.9734 (leading)

$$\text{Current } |I| = \frac{|V|}{|Z|}$$

$$|I| = \left|\frac{30\angle 0°}{6.8 - j1.6}\right|$$

I = 4.294 A

Average power,

$\therefore P_{avg} = |I|^2 Re(Z)$

$= (4.294)^2 \times 6.8$

P_{avg} = 125.38 W

Rack Your Brain

If a sinusoidal voltage is applied to a circuit, then the product of voltage and current is:

a) Apparent power
b) True power
c) Reactive power
d) Average power

Reactive power:

- As I^2R represents the power of resistance, similarly, I^2X represents the power of L or C.
- In a passive network that contains inductors and capacitors, or both, a portion of energy entering it during one cycle is stored, and then returned to the source. During the period of energy return, the power is negative.
- Thus, the power involved in this exchange is called reactive power. Although the net effect of reactive power is zero.
- Practically, the inductor and capacitor with ac supply do not consume any energy as they absorb energy for half-cycle and then return it in the next half-cycle. So, the net energy absorbed or power absorbed, would be zero.
- By defining reactive power, we have made power also as a complex number which is real for resistance and imaginary for inductance and capacitance, as shown below:

i.e., AC power: $P_{ac} = I^2Z$

for resistance: $Z = R$

$$P = I^2R$$

For inductance : $Z = j\omega L$

Power, $P = I^2(jX_L)$

$$P = jI^2X_L$$

For capacitance : $Z = \frac{1}{j\omega C}$

Power $= I^2\left(\frac{1X_C}{j}\right)$

$$P = -jI^2X_C$$

Note: The difference between the real power and apparent power is due to the phase shift provided by the inductor and capacitor.

So, we can say reactive power is the reason for the difference between real and apparent power.

Example 17:

If reactive power is hypothetical, then why it is defined?

Sol:

In ac analysis, we try to treat L and C just like R so we introduce the concept of impedance.

i.e., $Z_L = j\omega L$

and, $Z_C = \frac{1}{j\omega C}$

- But impedance is imaginary for L and C, which means it is mathematical, not real.

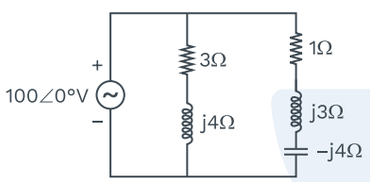
Now to compare L and C to R, we need to define power for L and C similar to R, and we call this power reactive power. But the reactive power is imaginary. As:

For the inductor, $Q_L = jI^2X_L$

For the capacitor, $Q_C = -jI^2X_C$

Example 18:

For the given circuit, calculate the reactive power absorbed by the circuit.



Sol:

The circuit can be redrawn as:

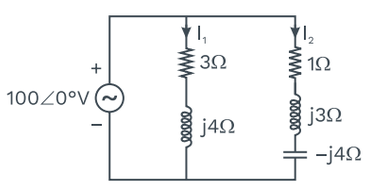


Fig. 7.23

From the circuit, $I_1 = \frac{V}{3 + j4} = \frac{100\angle 0°}{3 + j4}$

$$|I_1| = \frac{100}{5} = 20$$

and $I_2 = \frac{V}{1 + j3 - j4} = \frac{100\angle 0°}{1 - j}$

$$|I_2| = \frac{100}{\sqrt{2}}$$

So, reactive power:

$$Q = |I_1|^2 \times 4 + |I_2|^2 \times 3 - |I_2|^2 \times 4$$

$$Q = (20)^2 \times 4 + \left(\frac{100}{\sqrt{2}}\right)^2 \times 3 - \left(\frac{100}{\sqrt{2}}\right)^2 \times 4$$

$$Q = 400 \times 4 + 5000 \times 3 - 5000 \times 4$$

$$Q = 1600 + 15000 - 20000$$

$$Q = -3400 \text{ VAR}$$

Rack Your Brain

From the following options which is the most economical power factor for a consumer?

a) 0.25 – 0.5 lagging
b) 0.85 – 0.95 leading
c) 0.85 – 0.95 lagging
d) 0.25 – 0.5 leading

- Since the reactive power is negative which means the power is delivered not absorbed.

$\therefore$ Q = 3400 VAR

Complex power:

- Complex power is the product of the rms voltage phasor and the complex conjugate of the rms current phasor.
- In phasor form:

  $\vec{V} = V_{rms} \angle\theta_v$

  $\vec{I} = I_{rms} \angle\theta_i$

- If $\vec{I}^* = I_{rms} \angle -\theta_i$

then, $\vec{V}\vec{I}^* = V_{rms} I_{rms} \angle\theta_v - \theta_i$

- Real power or active power:

  $P = Re(\vec{V}\vec{I}^*)$

  $P = V_{rms} I_{rms} \cos(\theta_V - \theta_i)$ (Watts)
- Reactive power:

  $Q = Im(\vec{V}\vec{I}^*)$

  $Q = V_{rms} I_{rms} \sin(\theta_V - \theta_i)$ (VAR)

$\therefore$ Complex power,

$\bar{S}$ = Real or active + reactive power

$\bar{S} = V_{rms} I_{rms} \angle(\theta_v - \theta_i)$VA

$\bar{S} = P + jQ$

- Apparent power, $S = |\bar{S}| = V_{rms} I_{rms}$ (VA)
- As a complex quantity, its real part is real power P and its imaginary part is reactive power Q.
- The magnitude of complex power is called the apparent power.
- Complex power is the vector sum of active and reactive power.
- Complex power is measured in volt-amperes or VA to distinguish it from the average or real power, which is measured in watts.

Grey Matter Alert!

Complex power is important in the power analysis as it contains all the information on the power absorbed by the given load.

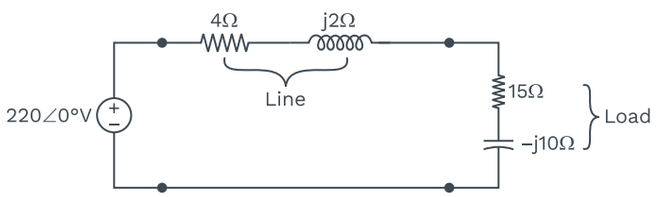
Rack Your Brain

The power factor for a purely inductive circuit is:

a) Unity
b) Lagging
c) Leading
d) None of the above

Example 19:

Calculate the complex power supplied by the source and complex power absorbed by the line and the load, as shown in the figure.



Sol:

From the above network,

4Ω and j2Ω are in series, and 15 Ω and –j10 Ω are in series, so impedance will be:

$Z_1 = (4 + j2)\Omega$

and $Z_2 = (15 - j10)\Omega$

Since, both the impedances are in series, so the equivalent impedance is:

$Z_{eq} = Z_1 + Z_2$

$= (4 + j2) + (15 - j10)$

$Z_{eq} = (19 - j8)\Omega$

We know,

$$\text{Current } |I| = \frac{|V|}{|Z_{eq}|} = \frac{220}{\sqrt{(19)^2 + 8^2}}$$

$$|I| = \frac{220}{\sqrt{425}} = \frac{220}{20.6}$$

$\vec{I} = 10.67\,A$

Complex power for line:

$\bar{S}_{line} = |I|^2 |Z_{line}|$

$= (10.67)^2 \sqrt{4^2 + 2^2}$

$\bar{S}_{line} = 509.33\,VA$

Complex power for load:

$\bar{S}_{load} = |I|^2 |Z_{load}|$

$= (10.67)^2 \sqrt{(15)^2 + 10^2}$

$\bar{S}_{load} = 2053.2\ VA$

Note: Apparent power can never be directly added or subtracted but complex power can be.

Grey Matter Alert!

The effects of low power factor are:

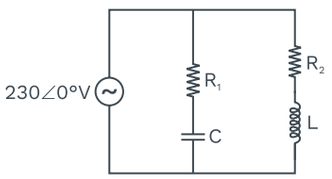
i) Large KVA rating of the equipment.

ii) Greater conductor size.

iii) Reduced handling capacity of the system.

Example 20:

A 230 V rms source supplies power to the two loads, which are connected in parallel. The first load draws 10 kW at 0.8 leading power factor and the second one draws 10 KVA at 0.8 lagging power factor. Calculate the complex power delivered by the source.



Sol:

We know whether the loads are connected in series or parallel, their complex power is added.

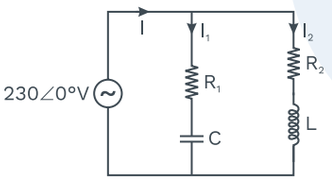


Fig. 7.24

According to the question,

For load 1:

Power = 10 kW

Power factor cos ϕ = 0.8 (leading)

So, power $P_1 = S_1 \cos \phi_1$

$$S_1 = \frac{P_1}{\cos \phi_1}$$

$$S_1 = \frac{10 \times 10^3}{0.8}$$

S_1 = 12500 VA

S_1 = 12.5 KVA

$\bar{S}_1 = 12.5 \angle -36.86° \text{KVA}$

{$\because$ Q<0 for leading pf}

For load 2:

S_2 = 10 KVA

Power factor, cos ϕ_2 = 0.8 (lagging)

$\bar{S}_2 = 10 \angle 36.86° \text{KVA}$ {$\because$ Q>0 for lagging pf}

So, the complex power is:

$\bar{S} = \bar{S}_1 + \bar{S}_2$

Rack Your Brain

The power factor of the system can be improved by installing the device in parallel with the load having:

a) Reactive power lagging

b) Reactive power leading

c) Apparent power

d) Resistive power

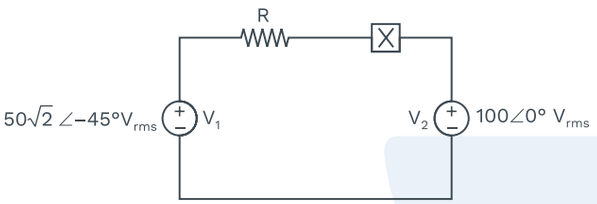
$= 12.5\angle -36.86° + 10\angle 36.86°$

$= (10 - j7.5) + (8 + j6)$

$\bar{S} = (18 - j1.5)\,VA$

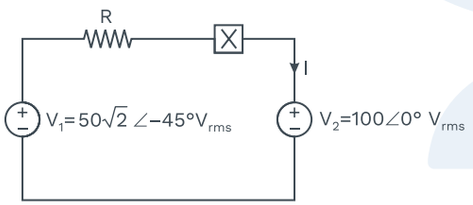
Example 21:

For the given circuit, the complex power absorbed by the source V_2 is (0 + j1000) VA. Calculate the value of R and unknown element X. (given $\omega = 10^4$ rad/sec)



Sol:

The circuit is redrawn as:



In the circuit, the current enters the positive terminal because the power is being absorbed.

Complex power is:

$S = V \cdot I^*$

$100\angle 0° \cdot I^* = 0 + j1000$

$\therefore I^* = j10$

$\therefore I = -j10$

or current:

$$I = \frac{V_1 - V_2}{Z}$$

$$-j10 = \frac{50\sqrt{2}\angle -45° - 100\angle 0°}{Z}$$

$$Z = \frac{50\sqrt{2}\angle -45° - 100\angle 0°}{-j10}$$

Grey Matter Alert!

Advantages of reactive power are:

- **i)** Improves the power factor of the system.
- **ii)** Reduces the losses in the network
- **iii)** Improves the system capacity
- **iv)** Improves the voltage regulation in the network
- **v)** Increases the power availability of the system

$$Z = \frac{\left[(50 - j50) - 100\right]}{-j10}$$

$$Z = \frac{-50 - j50}{-j10}$$

$$Z = (5 - j5)$$

$$\therefore R = 5\ \Omega$$

and (–j5) term means it is a capacitor,

$$\therefore X_C = 5 = \frac{1}{\omega C}$$

$$C = \frac{1}{\omega \times 5}$$

$$C = \frac{1}{10^4 \times 5}$$

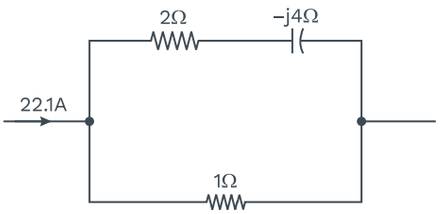
$$C = 20\ \mu F$$

Hence, $R = 5\Omega$

$X = C = 20\ \mu F$

Example 22:

For the parallel circuit shown in the figure, the complex power is:



Rack Your Brain

The reactive power (in MVAR) produced by a 10MVA generator operating at 0.86 power factor lagging is:

a) 5 **b)** 10

c) 8 **d)** 8.665

Sol:

In the given parallel circuit, 2Ω is in series with –j4Ω

So, $Z_1 = 2 + (-j4)\ \Omega$

$Z_1 = (2 - j4)\ \Omega$

and Z_1 is in parallel with 1Ω so equivalent impedance is:

$$Z_{eq} = Z_1 \parallel 1$$

$$= (2 - j4) \parallel 1$$

$$= \frac{(2 - j4) \times 1}{(2 - j4) + 1}$$

$$Z_{eq} = \frac{2 - j4}{3 - j4}$$

$$Z_{eq} = \frac{(2 - j4)(3 + j4)}{(3 - j4)(3 + j4)}$$

$$= \frac{(2 - j4)(3 + j4)}{9 + 16}$$

$$= \frac{(2 - j4)(3 + j4)}{25}$$

$$Z_{eq} = \frac{22 - j4}{25}$$

$$\vec{V} = \vec{I}\, Z_{eq}$$

and complex power is:

$$\vec{S} = \vec{V} \cdot \vec{I}^{*}$$

$$= \vec{I} \times Z_{eq} \times \vec{I}^{*} = |I|^2 Z_{eq}$$

$$= (22.1)^2 \times \frac{(22 - j4)}{25}$$

$$= 19.536\ (22 - j4)$$

$$\bar{S} = 429.8 - j78.144 \quad \text{(leading)}$$

$$S = \sqrt{(429.8)^2 + (78.144)^2}$$

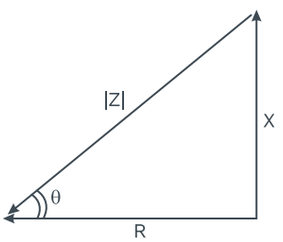
$$S = \sqrt{184728.04 + 6106.48}$$

$$S = 436.84\ \text{VA}$$

Impedance triangle and power triangle:

Impedance triangle:

- We know that impedance is:
  $Z = R + jX$
- Impedance triangle is a right-angled triangle whose base is resistance, perpendicular is reactance, and the hypotenuse is impedance.
- Impedance triangle is a geometrical representation of circuit impedance.



Where θ = impedance angle

$R = |Z| \cos\theta$

$X = |Z| \sin\theta$

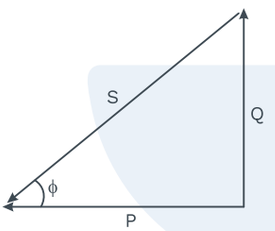
$Z = |Z| \angle\theta$

$= |Z| \cos\theta + j\,|Z| \sin\theta$

$Z = R + jX$

Power triangle:

- Power triangle is a right-angled triangle showing the relation between the active power, reactive power, and apparent power.



Power Triangle

Where

ϕ = power factor angle

$\phi = \theta$ (impedance angle)

$P = S \cos\phi$

$Q = S \sin\phi$

Power factor, $pf = \cos\phi = \frac{P}{S}$

Behavior of network elements:

1) Resistance:

- Voltage and current have the same phase



Fig. 7.25 Phasor Diagram of 'R' Element

$\therefore \quad \theta_v - \theta_i = 0$

$P = V_{rms}\, I_{rms} \cos(\theta_v - \theta_i)$

$P = V_{rms}\, I_{rms}$

$Q = V_{rms}\, I_{rms} \sin(\theta_v - \theta_i)$

$Q = 0$

$\therefore$ Power factor, $pf = \cos(\theta_v - \theta_i)$

$= \cos(0°)$

$\therefore \quad pf = 1$

- Resistance has unity power factor and it neither absorbs nor delivers reactive power(Q).

2) Inductance:

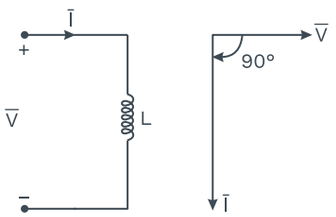


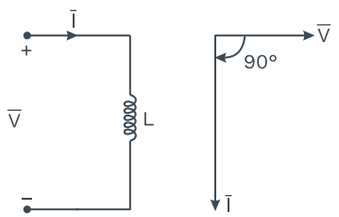
Fig. 7.26 L Circuit

- If current enters positive terminal, then,

$$v(t) = L\frac{di(t)}{dt}$$

$v(\omega) = j\omega L I(\omega)$

$\therefore$ $I(\omega)$ lags $V(\omega)$ by 90°



- By sign convention, L absorbs power

$\theta_v = 0°$, $\theta_i = -90°$

$\phi = \theta_v - \theta_I = 90°$

- Power factor pf = cos ϕ

pf = cos 90°

pf = 0 (lag)

$P = V_{rms}\, I_{rms} \cos(\theta_v - \theta_i)$

$P = V_{rms}\, I_{rms} \cos(90°)$

$P = 0$

$Q = V_{rms}\, I_{rms} \sin(\theta_v - \theta_i)$

$= V_{rms}\, I_{rms} \sin(0-(-90°))$

$Q = V_{rms}\, I_{rms}$

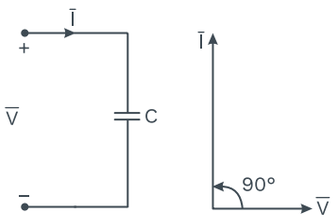
- Q > 0, the inductor does not absorb any real power but only absorbs reactive power.

Previous Years' Question

A 230 V rms source supplies power to two loads that are connected in parallel. The first load draws 10 kW at 0.8 leading power factor and the second one draws 10 kVA at 0.8 lagging power factor. Then the complex power delivered by the source is:

a) (18 + j1.5) kVA
b) (18 – j1.5) kVA
c) (20 + j1.5) kVA
d) (20 – j1.5) kVA

3) Capacitor:



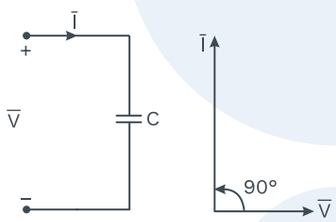
C Circuit

$$i(t) = C\frac{dv(t)}{dt}$$

$I(\omega) = j\omega CV(\omega)$

$I(\omega) = \omega CV(\omega) \angle 90°$

$I(\omega)$ leads $V(\omega)$ by 90°



- From the phasor diagram,
  $\theta_v = 0$
  $\theta_i = 90°$
- Power factor, pf = $\cos(\theta_v - \theta_i)$
  pf = $\cos(0 - 90°)$
  pf = 0° (lead)
- Power, P = $V_{rms}\ I_{rms} \cos(0 - 90°)$
  P = 0
  Q = $V_{rms}\ I_{rms} \sin(0 - 90°)$
  Q = $-V_{rms}\ I_{rms}$
- Since, by sign convention, C absorbs power but Q < 0. So, the capacitor does not absorb any power but delivers reactive power.

Example 23:

Determine the complete power triangle for the applied voltage, V = 240 sin(ωt – 60°)V and resulting current I = 13.33 cos(ωt + 48.7)A.

Sol:

Given, V = 240 sin (ωt – 60°)V

I = 13.33 cos(ωt + 48.7)A.

I = 13.33 sin (90° + ωt + 48.7°)

I = 13.33 sin (ωt + 138.7°)A

We know,

Real power, $P = V_{rms} \cdot I_{rms} \cos(\theta_v - \theta_i)$

$$P = \frac{240}{\sqrt{2}} \times \frac{13.33}{\sqrt{2}} \cos(-60° - 138.7°)$$

$$P = -1515.157 \text{ W}$$

Reactive power, $Q = V_{rms} \cdot I_{rms} \sin(\theta_V - \theta_i)$

$$Q = \frac{240}{\sqrt{2}} \times \frac{13.33}{\sqrt{2}} \times \sin(-60° - 138.7)$$

$$Q = 512.85 \text{ VAR}$$

Apparent power,

$$S = \sqrt{P^2 + Q^2}$$

$$S = \sqrt{(-1515.157)^2 + (512.85)^2}$$

$$S = 1599.59 \text{ VA}$$

Since real power is negative and reactive power is positive so the complete power triangle will be:

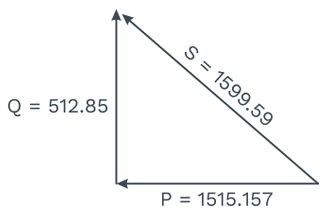
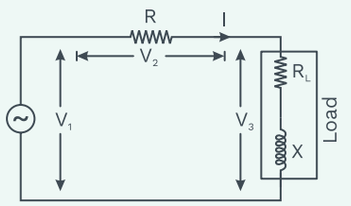


Fig. 7.27

Previous Years' Question

For the circuit shown below, the three voltmeter readings are $V_1 = 220$ V, $V_2 = 122$ V, $V_3 = 136$ V.



The power factor of the load is:

a) 0.45 **b)** 0.5

c) 0.55 **d)** 0.6

Power in harmonics:

- If,

$$v(t) = V_{m_1} \sin(\omega t + \theta_1) + V_{m_3} \sin(3\omega t + \theta_3) + V_{m_5} \sin(5\omega t + \theta_5)$$

and $i(t) = I_{m_1} \sin(\omega t + \phi_1) + I_{m_3} \sin(3\omega t + \phi_3)$

- Instantaneous power:

  $p(t) = v(t).i(t)$

This instantaneous power will contain 6 terms.

- Thus, the average power $P_{avg} = \frac{1}{T}\int_T p(t)d(t)$

- The terms which contain a product of two sinusoids of different frequencies will have integration equal to zero as the sinusoids of different frequencies are orthogonal to each other.

- Therefore, the average power would contain terms that are of the same frequency.

$$P = \frac{V_{m_1}}{\sqrt{2}} \cdot \frac{I_{m_1}}{\sqrt{2}} \cos(\theta_1 - \phi_1) + \frac{V_{m_3}}{\sqrt{2}} \cdot \frac{I_{m_3}}{\sqrt{2}} \cos(\theta_3 - \phi_3) \quad \text{....1)}$$

- Thus, this equation (1) is average power or real power.
- For apparent power:

$S = V_{rms}\, I_{rms}$

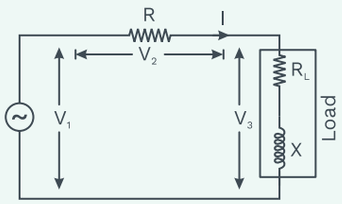
$$V_{rms} = \sqrt{\frac{1}{2}\left(V_{m_1}^2 + V_{m_3}^2 + V_{m_5}^2\right)}$$

and $$I_{rms} = \sqrt{\frac{1}{2}\left(I_{m_1}^2 + I_{m_3}^2\right)}$$

- Power factor = $\frac{P}{S}$

Previous Years' Question

In the circuit shown, the three voltmeter readings are V_1 = 220 V, V_2 = 122 V, V_3 = 136 V.



If R_L = 5 Ω, the approximate power consumption in the load is:

a) 700W **b)** 750W
c) 800W **d)** 850W

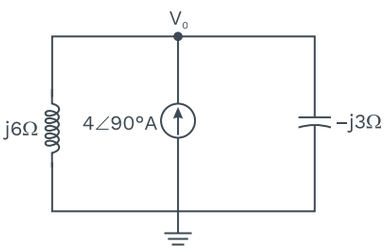
NODAL AND MESH ANALYSIS IN AC CIRCUITS

Nodal analysis:

- Nodal analysis for DC and AC circuits is the same. The only difference is the impedance in ac circuits rather than plain resistance in the DC circuits.
- It is one way to calculate the voltage and current distribution in ac circuits with harmonic/time-dependent sources. The basis of the nodal analysis is Kirchhoff's law. Since KCL is valid for the phasors, so we can analyze ac circuits by the nodal analysis.

Example 24:

Using the nodal analysis, find the value of V_0 in the circuit.



Sol:

Applying KCL in the circuit, we get:

$$\frac{V_0}{+j6} - \frac{V_0}{j3} = 4\angle 90°$$

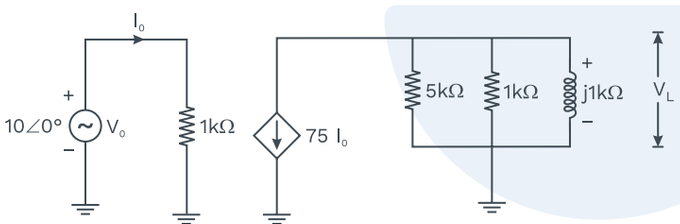
$$\frac{V_0 - 2V_0}{j6} = 4\angle 90°$$

$$-V_0 = 4j \times j6$$

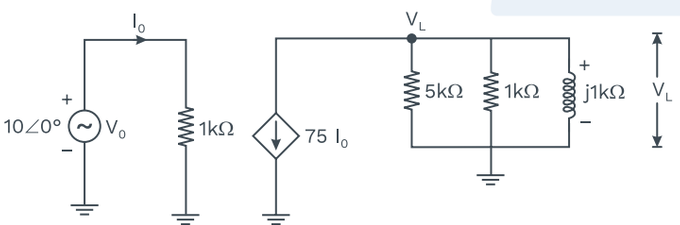
$$-V_0 = -24$$

$$\therefore V_0 = 24\ V$$

Example 25:
For the circuit shown below, obtain the voltage V_L across the inductor using nodal analysis.



Sol:
The circuit is given as:



So, from the above circuit, the current is:

$$I_0 = \frac{V_0}{1k} = \frac{10\angle 0°}{1 \times 10^3}$$

$$I_0 = 10\angle 0°\,mA$$

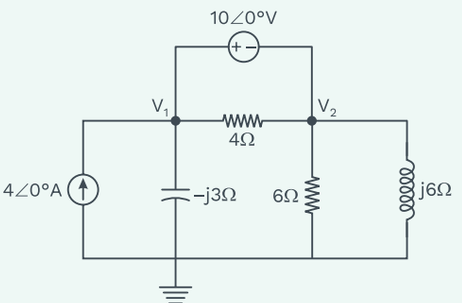
Applying nodal analysis at V_L:

$$\frac{V_L}{5k} + \frac{V_L}{1k} + \frac{V_L}{j1k} + 75I_0 = 0$$

$$\frac{V_L}{5 \times 10^3} + \frac{V_L}{1 \times 10^3} + \frac{V_L}{j1 \times 10^3} + 75 \times 10 \times 10^{-3}\angle 0° = 0$$

Previous Years' Question

For the circuit shown in the figure, the value of voltage V_2 (in volts) is:



a) 22 + j 2 **b)** 2 + j 22
c) 22 – j 2 **d)** 2 – j 22

$$\frac{V_L + 5V_L - j5V_L + 3750}{5 \times 10^3} = 0$$

$$6V_L - j5V_L + 3750 = 0$$

$$V_L(6 - j5) + 3750 = 0$$

$$V_L = \frac{-3750}{6 - j5}$$

$$V_L = 480 \angle -140.20° \text{ V}$$

Mesh analysis:

- Mesh analysis is a result of simplifying Kirchhoff's voltage law (KVL). For ac circuits, mesh analysis is the sum of impedances on the loop, multiplied by the loop current, minus the sum of impedances common to this loop, and the next loop is multiplied by the next loop, current which is equal to the sum of applied voltages to this loop.
- The only difference between the mesh analysis in dc circuits and ac circuits is that the ac circuits deal with impedances, i.e., capacitive, inductive, or resistive.

Example 26:

For the circuit shown, find the mesh currents I_1 and I_2 in the network.

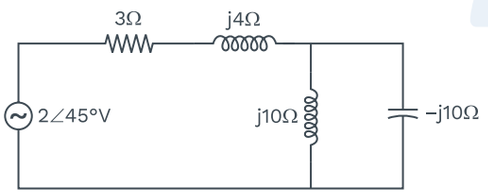


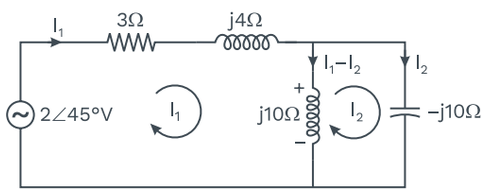
Fig. 7.28

Definitions

Impedance angle: The angle between the voltage and current is termed as impedance angle.

Sol:

The circuit can be redrawn as:



Applying KVL in 1st loop, we get:

$$2\angle 45° - I_1 \times 3 - I_1 \times j4 - j10(I_1 - I_2) = 0$$

$$2\angle 45° - 3I_1 - j4I_1 - j10I_1 + j10I_2 = 0$$

$$2\angle 45° - 3I_1 - j14I_1 + j10I_2 = 0$$

$$2\angle 45° = 3I_1 + j14I_1 - j10I_2$$

$$2\angle 45° = I_1(3 + j14) - j10I_2 \quad \ldots(1)$$

Applying KVL in 2nd loop, we get:

$$j10(I_1 - I_2) + j10I_2 = 0$$

$$j10I_1 - j10I_2 + j10I_2 = 0$$

$$j10I_1 = 0$$

$$\therefore I_1 = 0A$$

Putting value of I_1 in equation 1), we get:

$$2\angle 45° = 0(3 + j14) - j10I_2$$

$$2\angle 45° = -j10I_2$$

$$I_2 = \frac{2\angle 45°}{10\angle -90°}$$

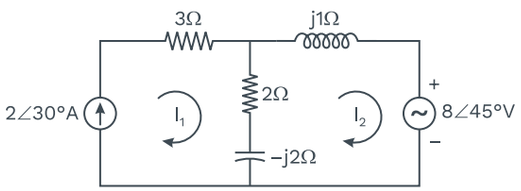
$$\therefore I_2 = 0.2\angle 135° \text{ A}$$

Definitions

A power triangle is a right triangle that defines the base angle to determine the cosine of that angle as the power factor (pf) of the load or the system. The size of the triangle determines the magnitudes of the power.

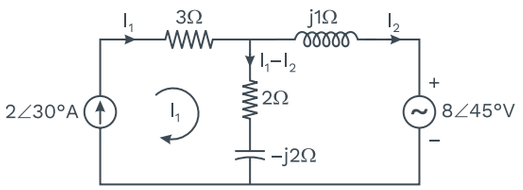
Example 27:

For the network shown below, determine the drop across 2Ω resistor using the mesh analysis.



Sol:

The circuit is redrawn as:



From the above circuit,

$$I_1 = 2\angle 30° \text{ A}$$

Applying KVL in the 2nd loop, we get:

$$2(I_1 - I_2) - j2(I_1 - I_2) - j1I_2 - 8\angle 45° = 0$$

$$(2 - j2)(I_1 - I_2) - j1I_2 - 8\angle 45° = 0$$

$$(2 - j2)I_1 - (2 - j2)I_2 - j1I_2 - 8\angle 45° = 0$$

$$(2 - j2)I_1 - 8\angle 45° = (2 - j2)I_2 + j1I_2$$

$$(2 - j2)2\angle 30° - 8\angle 45° = (2 - j2 + j1)I_2$$

$$I_2 = \frac{(2 - j2)2\angle 30° - 8\angle 45°}{(2 - j1)}$$

$$I_2 = 3.184 \angle -64.985° \text{ A}$$

Now, the voltage across 2Ω resistor is:

$$V_{2\Omega} = (I_1 - I_2) \times 2$$

$$= [2\angle 30° - 3.184\angle -64.985] \times 2$$

$$V_{2\Omega} = 7.81\angle 84.33° \text{ V}$$

Grey Matter Alert!

The phase shift between $\vec{V}$ and $\vec{I}$ is produced due to the inductor and capacitor. No phase shift occurs due to resistor.

Network theorems in AC circuits:

Superposition theorem:

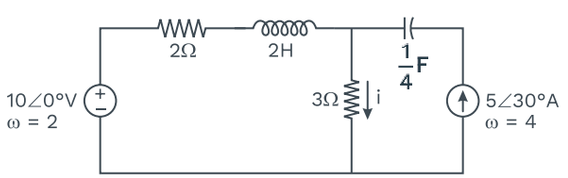
- Superposition theorem is very important in ac analysis because we may have sources of different frequencies in a network. As the frequency of the source changes, the reactance offered by the inductor and the capacitor also changes, which means the entire circuit will be different at a different frequency.
- So, we consider one source at a time and draw the entire network based on the frequency of the source and analyze one source at a time. The responses obtained for each source must not be added as phasors, as each one has a different frequency, so we need to convert them to the time domain and express the result as the sum of these responses.

For example:

$$i(t) = I_{m_1} \sin(\omega_1 t + \phi_1) + I_{m_2} \cos(\omega_2 t + \phi_2)$$

Example 28:

For the circuit shown in the figure, find the current i?



Grey Matter Alert!

Superposition theorem eliminated the need for solving the simultaneous linear equations by considering the effect of each source independently.

Sol:

In the network given, firstly, we consider the source of $\omega = 2$.

Reactance of 2H inductor,

$$X_L = \omega L = 2 \times 2$$

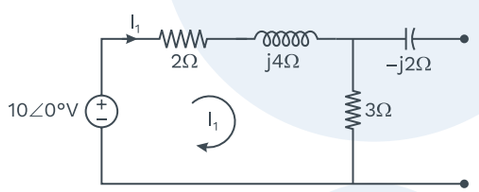
$$X_L = 4\Omega$$

Reactance of $\frac{1}{4}$F capacitor,

$$X_C = \frac{1}{\omega C} = \frac{1}{2 \times \frac{1}{4}}$$

$$X_C = 2$$

Now, the current source is open-circuited from the network, then the circuit becomes:



Current, $I_1 = \frac{V}{2 + j4 + 3}$

$$I_1 = \frac{10\angle 0°}{5 + j4}$$

$$I_1 = \frac{10}{\sqrt{41}} \angle - \tan^{-1} 0.8$$

$$I_1 = 1.56 \angle - 38.66° \text{ A}$$

Now considering the current source and the voltage source will be short-circuited.

$\because \omega = 4$

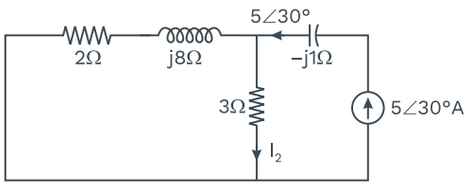
$$X_L = 4 \times 2$$

$$X_L = 8\ \Omega$$

$$X_C = \frac{1}{4 \times \frac{1}{4}}$$

$$X_C = 1\ \Omega$$

The circuit becomes:



The current I_2 in the circuit is:

$$I_2 = 5\angle 30° \frac{(2 + j8)}{(5 + j8)}$$

$$I_2 = 4.37\angle 47.97° \text{ A}$$

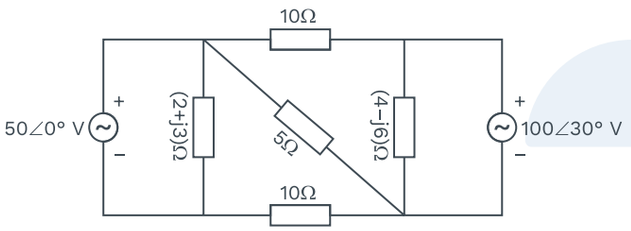
$$i(t) = 1.56\sin(2t - 38.66°) + 4.37\sin(4t + 47.97°)\text{A}$$

Rack Your Brain

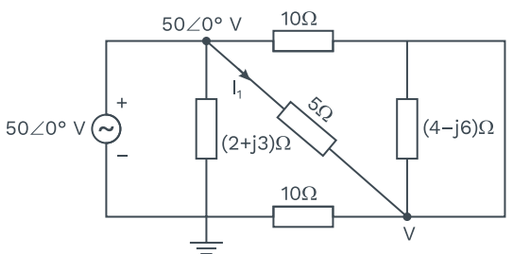
In the power triangle, one of the sides is representing the apparent power (S) lies in the 1st quadrant, then the load power factor is:

a) Unity **b)** Leading
c) Lagging **d)** Zero

Example 29:
Using the superposition theorem, find the current through 5Ω resistance.



Sol:
In the given network, considering $50\angle 0°\,V$ as the source, so the other voltage will be short-circuited as shown below,



Applying nodal analysis at V:

$$\frac{V - 0}{10} + \frac{V - 50\angle 0°}{5} + \frac{V - 50\angle 0°}{10} = 0$$

$$\frac{V + 2V - 100 + V - 50}{10} = 0$$

$$\frac{4V - 150}{10} = 0$$

$$4V - 150 = 0$$

$$4V = 150$$

$$V = \frac{150}{4}$$

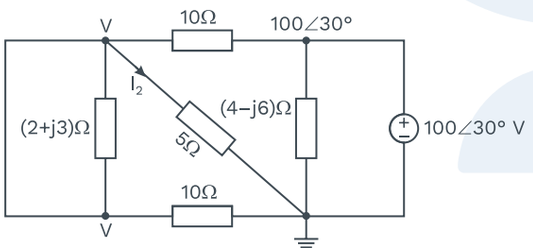
$$V = 37.5 \text{ V}$$

Current,

$$I_1 = \frac{50\angle 0° - 37.5}{5}$$

$$I_1 = 2.5 \text{ A}$$

Now considering $100\angle 30° \text{ V}$ source and other voltage sources will be short-circuited, as shown in the figure below:



Applying nodal at V node, we get:

$$\frac{V - 0}{5} + \frac{V - 100\angle 30°}{10} + \frac{V - 0}{10} = 0$$

$$\frac{2V + V - 100\angle 30° + V}{10} = 0$$

$$\frac{4V - 100\angle 30°}{10} = 0$$

$$4V - 100\angle 30° = 0$$

$$4V = 100\angle 30°$$

$$V = 25\angle 30° \text{ V}$$

$$\therefore I_2 = \frac{25\angle 30° - 0}{5}$$

$$\therefore I_2 = 5\angle 30° \text{ A}$$

Thus, the total current across the 5Ω resistor is:

Grey Matter Alert!

For the symmetrical waveforms, only "odd" harmonics may appear in multiples of 3rd, 5th, 7th, etc., of the fundamental frequency. For asymmetrical waveforms, a part from "odd", "even" multiples of the fundamental may appear in multiples 2nd, 4th, 6th, etc., Also DC components appear in asymmetrical waveforms, which are represented as 0Hz signals.

$\vec{I} = \vec{I}_1 + \vec{I}_2$

$\vec{I} = 2.5 + 5\angle 30°$

$\therefore I = 7.27\angle 20.1°$ A

Thevenin's theorem:

- A linear active R-L-C network that contains one or more independent or dependent voltage and current sources can be replaced by a single voltage source V_{OC} in series with equivalent impedances.

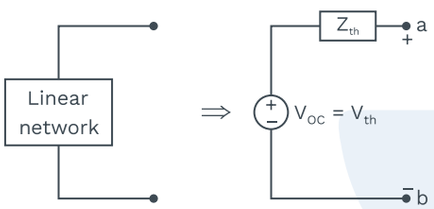
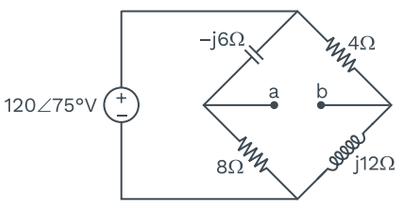


Fig. 7.29 Thevenin equivalent circuit

- In ac circuit, nothing changes as compared to DC circuit except resistance gets replaced by the impedance, and voltage and current becomes phasor.

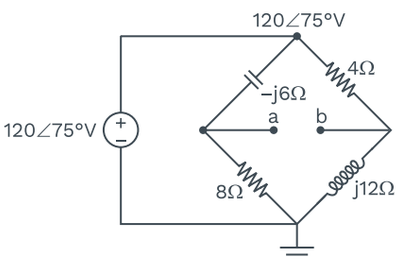
Example 30:

Determine the Thevenin's equivalent at terminal a-b of the circuit.



Sol:

The circuit can be redrawn as:



Grey Matter Alert!

The supernode concept is used instead of nodal analysis when a voltage source (dependent or independent) is connected between the two non-reference nodes.

By voltage division rule:

$$V_a = 120\angle 75° \times \frac{8}{8 - j6}$$

$$V_a = \frac{960\angle 75°}{10\angle -36.86°}$$

$$V_a = 96\angle 111.86° \text{ V}$$

$$V_b = 120\angle 75° \times \frac{j12}{4 + j12}$$

$$V_b = \frac{14400\angle 165°}{\sqrt{160}\angle \tan^{-1} 3}$$

$$V_b = 113.84\angle 93.435° \text{ V}$$

Thevenin's voltage V_{Th} is:

$$V_{Th} = V_b - V_a$$

$$= 113.84\angle 93.435° - 96\angle 111.86°$$

$$V_{th} = 37.93\angle 40.31°$$

For Z_{Th} short circuit the voltage source.

$$Z_{th} = [(8)||(-j6)] + [(j12)||(4)]$$

$$Z_{th} = \frac{-j48}{8 - j6} + \frac{j48}{4 + j12}$$

$$Z_{th} = 7\angle -22.16° \Omega$$

Norton's theorem:

- A linear active RLC network that contains one or more independent voltage and current sources can be replaced by a single current source in shunt with an equivalent impedance.

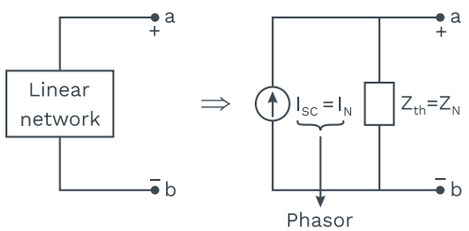


Fig. 7.30 Norton Equivalent Circuit

$$I_N = \frac{V_{th}}{Z_{th}}$$

$$Z_N = Z_{th}$$

Grey Matter Alert!

Advantages of Thevenin's theorem are:

i) It reduces a complex circuit to a simple circuit by a single source of emf in series with a single resistance.

ii) It simplifies the portion of the circuit of lesser impedance and enables us to view the action of the output part directly.

iii) This theorem is useful to find the current in a particular branch of the network as that resistance of the branch is varied while all other resistances and emf source remains constant.

Rack Your Brain

In a circuit, if there are N nodes present, then the number of nodal equations that can be formed are?

a) N-1 **b)** N

c) N+1 **d)** N-2

Example 31:
Obtain the current through the 1Ω resistor for the figure shown below using Norton's theorem.

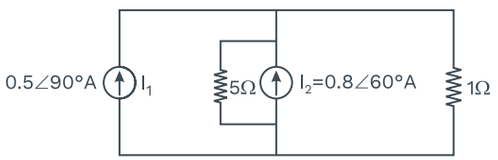
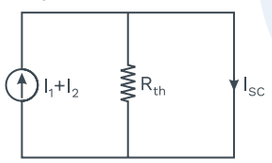


Fig. 7.31 Maximum power Transfer circuit

Sol:
For Norton's theorem, both the current I_1 and I_2 will be added up. Thus, the circuit becomes



Currents, $I_1 = 0.5\angle 90° \text{ A} = j0.5 \text{ A}$

$I_2 = 0.8\angle 60° \text{ A} = 0.4 + j0.4\sqrt{3} \text{ A}$

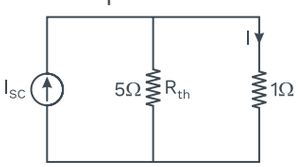
$I_{SC} = I_1 + I_2$

$= j0.5 + 0.4 + j0.4\sqrt{3}$

$I_{SC} = 0.4 + j\left(0.5 + 0.4\sqrt{3}\right)$

and $R_{th} = 5\ \Omega$

Thus, the Norton equivalent is:



$I = I_{SC} \times \left(\frac{5}{6}\right)$

$I = \frac{2}{6} + j\frac{5}{6}\left(0.5 + 0.4\sqrt{3}\right)$

$I = 0.37\angle 25.736° \text{ A}$

Grey Matter Alert!

Limitations of Norton's theorem are as follows:

i) Norton's theorem is valid only for the linear elements and invalid for non-linear elements.

ii) Norton's theorem does not apply to the circuits consisting of unilateral elements.

iii) Norton's theorem does not apply to the circuits consisting of a load in series or parallel with controlled sources or dependent sources.

iv) Norton's theorem does not apply to the circuits consisting of magnetic locking, coupling between the load, or any other circuit element.

v) The power dissipation across Norton's equivalent circuit is not identical with the power dissipation in the real circuit.

Maximum power transfer theorem:

- In ac circuits, we try to maximize the real power being delivered to the load, and hence maximum power transfer theorem is applied to real power.

- Before applying the maximum power transfer theorem, we need to determine the Thevenin equivalent of the circuit across a load.
- For the maximum power transfer theorem to be applied, load impedance must be variable.

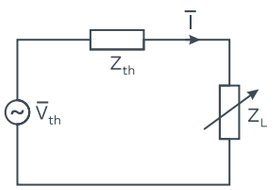


Fig. 7.32 Maximum power Transfer circuit

$$Z_{th} = R_{th} + jX_{th}$$

$$Z_L = R_L + jX_L$$

$$\text{Current, } \bar{I} = \frac{\bar{V}_{th}}{Z_{th} + Z_L}$$

$$= \frac{\bar{V}_{th}}{(R_{th} + R_L) + j(X_{th} + X_L)}$$

$$\text{Real power in load, } P_L = |I|^2 R_L$$

$$= \frac{|V_{th}|^2 R_L}{(R_{th} + R_L)^2 + (X_{th} + X_L)^2}$$

To maximize P_L,

$$\frac{dP_L}{dR_L} = 0$$

$$0 = \frac{|V_{th}|^2 \left[\left\{(R_{th} + R_L)^2 + (X_{th} + X_L)^2\right\} - R_L\left\{2(R_{th} + R_L)\right\}\right]}{\left[(R_{th} + R_L)^2 + (X_{th} + X_L)^2\right]^2}$$

$$(R_{th} + R_L)\,[R_{th} + R_L - 2R_L] + (X_{th} + X_L)^2 = 0$$

$$\therefore R_L = R_{th} \text{ and } X_L = -X_{th}$$

$$\frac{dP_L}{dX_L} = 0$$

$$0 = \frac{-|V_{th}|^2 R_L \times 2(X_{th} + X_L)}{\left[(R_{th} + R_L)^2 + (X_{th} + X_L)^2\right]^2}$$

$$X_{th} + X_L = 0$$

$$X_L = -X_{th}$$

$$\text{So, } R_L = R_{th}$$

Rack Your Brain

Mesh analysis is used to obtain:

a) Current **b)** Voltage
c) Power **d)** Resistance

$\therefore Z_L = R_L + jX_L = R_{th} - jX_{th}$

$\therefore Z_L = Z_{th}^*$

Note: There are only 3 types of power in circuits, the power which is consumable or convertible is active power.
So, the maximum power transfer theorem in ac circuits is confined to real power only.

Cases for maximum power transfer theorem:

Case 1: General source and general load with Z_L = variable.

$Z_L = R_L + jX_L$

$Z_L = Z_{th}^*$

$R_L = R_{th}$

$X_L = -X_{th}$

$$\therefore I = \frac{V_{th}}{(R_{th} + R_L) + j(X_{th} + X_L)}$$

$$I = \frac{V_{th}}{2R_{th}}$$

So, maximum power, $P_{max} = |I|^2 R_L$

$$= \frac{V_{th}^2}{4R_{th}^2} \cdot R_{th}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

Same as for dc circuits.

Case 2: General source and resistive load, i.e., R_L = variable.

$Z_L = R_L$

$X_L = 0$

$$\therefore \frac{dP_L}{dX_L} \text{ is invalid}$$

$$\therefore \frac{dP_L}{dR_L} = 0$$

$(R_{th} + R_L)(R_{th} - R_L) + (X_{th} + X_L)^2 = 0$

$R_{th}^2 - R_L^2 + X_{th}^2 = 0$

$$\therefore R_L = \sqrt{R_{th}^2 + X_{th}^2}$$

$R_L = |Z_{th}|$ for maximum power transfer theorem

Rack Your Brain

For a non-ideal voltage source V_S has an internal impedance Z_S. If a purely resistive load is chosen that maximized the power transferred to the load, then the value must be:

a) Complex conjugate of Z_S
b) Magnitude of Z_S
c) Real part of Z_S
d) Zero

$$I = \frac{V_{th}}{(R_L + R_{th}) + jX_{th}}$$

$$I = \frac{V_{th}}{|Z_{th}| + R_{th} + jX_{th}}$$

Maximum power is:

$$P_{max} = |I|^2 R_L$$

$$P_{max} = |I|^2 |Z_{th}|$$

Note: In this case, do not use $\frac{V_{th}^2}{4R_{th}}$ directly instead compute current and then power.

Case 3: General source and general load with R_L as variable.

$$Z_L = R_L + jX_L$$

Where, R_L = variable
and X_L = fixed
$\therefore \frac{dP_L}{dX_L}$ will be invalid

So, $\frac{dP_L}{dR_L} = (R_{th} + R_L)(R_{th} - R_L) + (X_{th} + X_L)^2 = 0$

$$R_{th}^2 - R_L^2 + (X_{th} + X_L)^2 = 0$$

$$R_L = \sqrt{R_{th}^2 + (X_{th} + X_L)^2}$$

$$R_L = |Z|$$

Where Z is the total impedance excluding R_L.
So, to compute P_{max} find the current.

$$\therefore P_{max} = |I^2| R_L$$

Case 4: General source and general load with X_L as the variable.

$$Z_L = R_L + jX_L$$

Where, R_L = fixed
and, X_L = variable
$\frac{dP_L}{dR_L}$ will be invalid

So, $\frac{dP_L}{dX_L} = 0$

$$X_{th} + X_L = 0$$

$$X_L = -X_{th}$$

Rack Your Brain

The superposition theorem is used to compute:
a) Power
b) Energy
c) Resistance
d) Impedance

Grey Matter Alert!

Applications of maximum power transfer are:
i) In radio communication
ii) In the communication system
iii) In-car engines

Rack Your Brain

When the power transferred to the load is maximum then the efficiency of the power transfer is:
a) 75% **b)** 50%
c) 25% **d)** 100%

Current, $I = \dfrac{V_{th}}{(R_{th} + R_L) + j(X_{th} + X_L)}$

$I = \dfrac{V_{th}}{(R_{th} + R_L)}$

$\therefore P_{max} = |I|^2 \cdot R_L$

Case 5: Resistive source and resistive load with R_L as variable

$Z_{th} = R_{th}$

$Z_L = R_L$

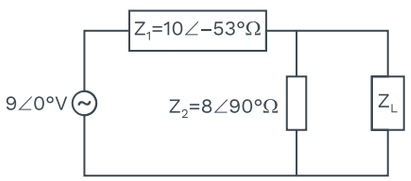
$\dfrac{dP_L}{dR_L} = 0$

$R_L = R_{th}$

$P_{max} = \dfrac{V_{th}^2}{4R_{th}}$ [same as dc circuit]

Example 32:

For the network shown, find the value of Z_L for maximum power.



Sol:

We know from the maximum power transfer theorem,

$Z_L = Z_{th}^*$

Z_{th} is the Thevenin impedance across Z_L

Now, Z_L will be an open circuit, and the source voltage will be a short circuit as shown in the figure.

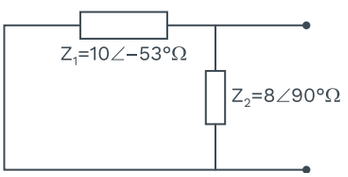


Fig. 7.33

$\therefore Z_{th} = Z_1 \parallel Z_2$

$= 10\angle -53° \parallel 8\angle 90°$

$= (6 - j8) \parallel j8$

Rack Your Brain

The superposition theorem is only applicable to:

a) Non-linear elements

b) Resistive elements

c) Passive elements

d) Linear bilateral elements

$$= \frac{(6 - j8) \times j8}{6 - j8 + j8}$$

$$Z_{th} = \frac{64 + j48}{6}$$

$$Z_{th} = (10.66 + j8)\Omega$$

$$Z_L = Z_{th}^{*} = (10.66 - j8)\Omega$$

Example 33:

In the given network, for the maximum power dissipation in R. Calculate the value of R.

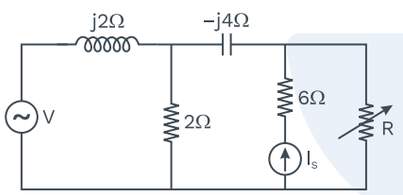


Fig. 7.34

Rack Your Brain

Is the superposition theorem applicable to power?

Sol:

For the calculation of Z_{th}, the voltage source will be short-circuited, and the current source will be open-circuited.

So, $Z_{th} = (2 \parallel j2) - j4$

$$Z_{th} = \frac{2 \times j2}{2 + j2} - j4$$

$$Z_{th} = \frac{j4}{2 + j2} - j4$$

$$Z_{th} = \frac{j2(1 - j)}{2} - j4$$

$$Z_{th} = 1 + j - j4$$

$$Z_{th} = 1 - j3$$

We know that for the maximum power theorem,

$$R = |Z_{th}|$$

$$R = \sqrt{(1)^2 + (3)^2}$$

$$R = \sqrt{10}$$

$$\therefore R = 3.16\,\Omega$$

5) Millman's theorem:

Millman's theorem for ac circuits is the same as in dc circuits, the only difference is that the resistance R is replaced by impedance Z.

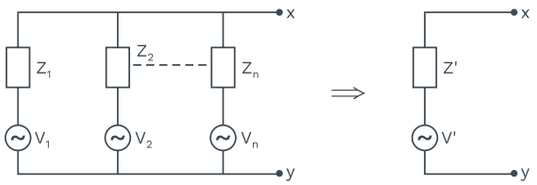


Fig. 7.35 Millman's Theorem

$$\therefore V' = \frac{\sum_{i=1}^{n} \frac{V_i}{Z_i}}{\sum_{i=1}^{n} \frac{1}{Z_i}} \text{ and } Z' = \frac{1}{\sum_{i=1}^{n} \frac{1}{Z_i}}$$

Example 34:

Using Millman's theorem, find the value of current I.

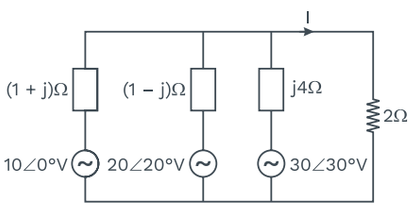


Fig. 7.36

Grey Matter Alert!

Millman's theorem was named after the famous electrical engineering professor JACOB MILLMAN, who proposed the idea of this theorem. This theorem is a combination of Thevenin's Theorem and Norton's Theorem. This theorem is used to find out the voltage across the load and the current through the load.

Sol:

By Millman's theorem,

$$V' = \frac{\sum_{i=1}^{n} \frac{V_i}{Z_i}}{\sum_{i=1}^{n} \frac{1}{Z_i}}$$

$$\therefore V' = \frac{\frac{10\angle 0°}{1+j} + \frac{20\angle 20°}{1-j} + \frac{30\angle 30°}{j4}}{\frac{1}{1+j} + \frac{1}{1-j} + \frac{1}{j4}}$$

$$V' = 14.34\angle 19.158° \text{ V}$$

$$Z' = \frac{1}{\sum_{i=1}^{n} \frac{1}{Z_i}} = \frac{1}{\frac{1}{1+j} + \frac{1}{1-j} + \frac{1}{j4}}$$

$$\therefore Z' = \left(\frac{1 + j0.25}{1.0625}\right)\Omega$$

$$I = \frac{V'}{2 + Z'}$$

$$I = \frac{14.34\angle 19.16°}{2 + \left(\frac{1 + j0.25}{1.0625}\right)}$$

$$I = 4.86\angle 14.59° \text{ A}$$

Chapter summary



- The final output of a system when transient has died out is called a steady-state response.
- The opposition offered by a network to the flow of ac current is termed as impedance.

$$Z = \frac{V}{I}\Omega$$

- Admittance represents the allowance to the flow of ac current.

$$Y = \frac{1}{Z} = \frac{I}{V}\mho$$

- Phasor is a vector representation of the sinusoidal signal to simplify the algebraic operations on sinusoidal quantity.
- Phasor diagram for a purely resistive load with voltage as a reference is-



Fig. 7.37 Phasor Diagram of R Element

For purely resistive load, Impedance angle $\theta = 0°$
Power factor angle $\phi = 0°$
The current is in phase with the voltage.

- The impedance angle for purely inductive load is $\theta = 90°$
  The power factor angle for purely inductive load is $\phi = -90°$
  Current lags voltage by an angle of 90°.
  Phasor diagram for purely inductive load with voltage as a reference is given by,

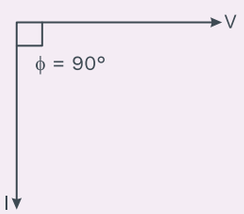


Fig. 7.38 Phasor Diagram for 'L' Load

- The impedance angle for purely capacitive load is $\theta = -90°$
  The power factor angle for purely capacitive load is $\phi = 90°$
  Current leads voltage by an angle of 90°.
  Phasor diagram for purely capacitive load with voltage as a reference is given by,

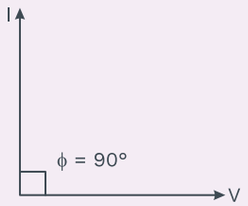


Fig. 7.39 Phasor Diagram for 'C' Load

- Instantaneous power is defined as the product of the instantaneous terminal voltage and current.

$$p(t) = V_{rms}\, I_{rms}\left[\cos(2\omega t + \theta_v + \theta_i) + \cos(\theta_v - \theta_i)\right]$$

- Average power is defined as the average of the instantaneous power over one period.

$$P_{avg} = \frac{1}{T}\int_0^T p(t)\,dt$$

- Average power (active power) consumed by purely resistive load is given by,

$$P_{avg} = \frac{1}{T}\int_0^T (V_m \sin\omega t)(I_m \sin\omega t)\,d\omega t$$

$$P_{avg} = \frac{V_m I_m}{2}$$

- Average power (active power) consumed by purely inductive load is given by,

$$P_{avg} = \frac{1}{T}\int_0^T (V_m \sin\omega t) I_m \sin(\omega t - 90°)\,d\omega t$$

$$P_{avg} = 0$$

- Average power (active power) consumed by purely capacitive load is given by,

$$P_{avg} = \frac{1}{T}\int_0^T (V_m \sin\omega t) I_m \sin(\omega t + 90°)\,d\omega t$$

$$P_{avg} = 0$$

- Reactive power is the power consumed in an ac circuit that does not perform any useful work but has a great effect on the phase shift between the voltage and current waveform,
- Reactive power consumed by a purely resistive load is given by,

$$Q_{avg} = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \sin(\theta_v - \theta_i) = \frac{V_m I_m}{2} \sin(0) \quad (\because\ \theta_v - \theta_i = 0)$$

$Q_{avg} = 0$

- Reactive power consumed by a purely inductive load is given by,

$$Q_{avg} = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \sin(\theta_v - \theta_i) = \frac{V_m I_m}{2} \sin(0 - (-90°)) \quad \{\because \theta_v = 0, \theta_i = -90°\}$$

$$Q_{avg} = \frac{V_m I_m}{2} (\text{VAR})$$

- Reactive power consume by purely capacitive load is given by,

$$Q_{avg} = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \sin(\theta_v - \theta_i) = \frac{V_m I_m}{2} \sin(0 - 90°) \quad \{\because \theta_v = 0, \theta_i = 90°\}$$

$$Q_{avg} = -\frac{V_m I_m}{2} (\text{VAR})$$

- Apparent power is the product of the root mean square (RMS) value of voltage and current.
- The active power is the real power that is dissipated in the circuit.
  For resistance, $P = V_{rms} \cdot I_{rms}$

  For inductance, P = 0
  For capacitance, P = 0
- The ratio of real power to the apparent power is called as power factor,

$$pf = \cos(\theta_v - \theta_i)$$

  In resistance, $\vec{V}$ and $\vec{I}$ lies in the same phase.

  Power factor, pf = 1

  In inductance, $\vec{I}$ lags $\vec{V}$ by 90°.

  Power factor, pf = 0 (lagging)

  In capacitance, $\vec{I}$ leads $\vec{V}$ by 90°

  Power factor, pf = 0 (leading)
- Complex power is the vector sum of active and reactive power.

$$\bar{S} = V_{rms} I_{rms} \angle(\theta_v - \theta_i) \text{VA}$$

- The angle between the voltage and current is termed as impedance angle.
- A power triangle is a right triangle that defines the base angle to determine the cosine of that angle as the power factor (pf) of the load or the system.

PRACTICE QUESTIONS

Q1 **For the circuit shown below, the value of current I_C and I is:**

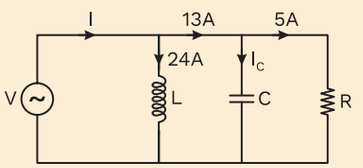


Fig. 7.40

a) I_C = 12A, I = 13A
b) I_C = 10A, I = 15A
c) I_C = 12A, I = 16A
d) I_C = 10A, I = 13A

Q2 **For the circuit shown below, the value of current $i_X(t)$ when the value of source voltage $V_S = 10\sin\omega t$ V is:**

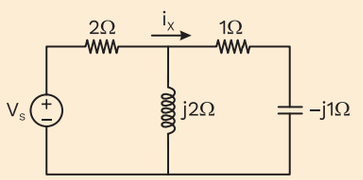


Fig. 7.41

Q3 **For RL series circuit R = 20 Ω and L = 20 mH, have an impedance of $40\angle\theta$ Ω. Then the angle θ and the frequency of the circuit are:**
a) 60°, 276 Hz
b) 120°, 251 Hz
c) 45°, 157 Hz
d) 120°, 276 Hz

Q4 **For a series RLC circuit, R = 15 Ω, L = 80mH, and C = 30μF, has a sinusoidal current at angular frequency 500 rad/s. then the phase angle of the circuit is:**
a) 58.5°
b) 60.6°
c) 23.1°
d) 66.0°

Q5 **For a network, the phasor voltage V across the terminals is $100\angle45°$V, and the resulting current I is $5\angle15°$A. Calculate the equivalent impedance and admittance.**

Q6 **The impedance of the circuit is Z = (8 – j6)Ω has an applied phasor voltage 70.7∠–90° V. Calculate the complete power triangle.**

a) P = 150 W, Q = 150 VAR (capacitive), pf = 0.866
b) P = 200 W, Q = 180 VAR (capacitive), pf = 0.866
c) P = 200 W, Q = 150 VAR (capacitive), pf = 0.80
d) P = 15 0W, Q = 180 VAR (capacitive), pf = 0.80

Q7 **By using the phasor approach, calculate the voltage v(t) represented by the integrodifferential equation.**

$$2\frac{dv}{dt} + 5v + 10\int v\,dt = 20\cos\left(5t - 30°\right)$$

a) 2.12 cos(t-88°) V **b) 2.12 cos(5t-88°) V**
c) 1.011 cos(5t-88°) V **d) 1.011 cos(t-88°) V**

Q8 **A circuit with the three impedances Z_1 = 25∠15°Ω, Z_2 = 15∠60°Ω, and Z_3 = 15∠90°Ω, has an applied voltage V_S = 339.4∠–30° V; find the total apparent power (in VA) of the circuit.**

Q9 **A network is shown in the figure; calculate the current in (2 + j3) Ω impedance due to the sources V_1 and V_2, respectively, by using the mesh current analysis.**

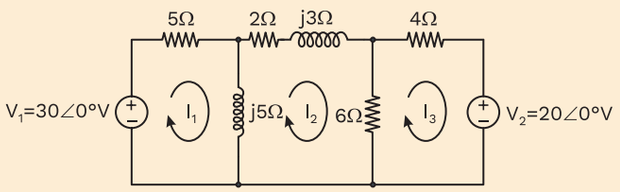


Fig. 7.42

Q10 **In the given network, find the total current in load Z_L using Millman's theorem.**

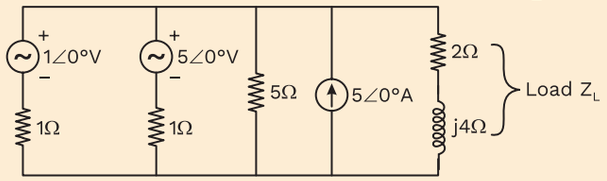


Fig. 7.43

Answer Key:

1) a)
2) $I_x = 3.53\sin(\omega t + 45°)$ V
3) a)
4) b)
5) $Z = 17.32 + j10.0\Omega$ and
   $Y = (4.33 - j2.50) \times 10^{-2}$ S
6) c)
7) b)
8) 4291
9) $2.41\angle 6.4°$A and $1.36\angle 141.4°$A
10) $I_L = 1.06\angle -58.465°$ A